Perfectoids and Galois cohomology: a pedagogical introduction to p-adic Hodge theory

何遇方

Lecture 1. §1 A glance at p-adic Hodge theory

§1.1 Hodge decomposition

Th. X projective smooth variety $/\mathbb{C}$

$$\underset{\text{singular coh}}{H^n_{sing}(X,\mathbb{C})} \simeq \bigoplus_{i+j=n} \underset{\text{coherent cohomology}}{H^j(X, \Omega^i_{X/\mathbb{C}})}$$

The proof uses techniques in analysis.

every (i,j)-cohomology class is represented by a harmonic diff. form.

§1.2 $\mathbb{C}$ and $\mathbb{C}_p$

$$\begin{tikzcd}
 & \mathbb{R} \arrow[r, "\text{alg. closure}"] & \mathbb{C} \arrow[r, "\text{completion}"] & \mathbb{C} \\
\mathbb{Q} \arrow[ur, "\text{arch. norm}"] \arrow[dr, "\text{na norm}"'] & & & \\
 & \mathbb{Q}_p \arrow[r, "\text{alg. closure}"] & \overline{\mathbb{Q}}_p \arrow[r, "\text{completion}"] & \mathbb{C}_p = \widehat{\overline{\mathbb{Q}}_p}
\end{tikzcd}$$

(p-adic analogue of $\mathbb{C}$)

complete alg. closed val. field

(Ostrovski)

$\mathbb{Q}_p = \mathbb{Z}_p[\frac{1}{p}]$ discrete valuation field.

$\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ DVR

$v_p : \mathbb{Z}_p \to \mathbb{N} \cup \{\infty\}$

$p^n u \mapsto n$, $u \in \mathbb{Z}_p^\times$

$0 \mapsto \infty$

$\overline{\mathbb{Z}}_p$ = int. closure of $\mathbb{Z}_p$ in $\overline{\mathbb{Q}}_p$ is valuation ring (non-discrete)

$v_p : \overline{\mathbb{Z}}_p \to \mathbb{Q}_{\geq 0} \cup \{\infty\}$

$p^{\frac{1}{n}} \mapsto \frac{1}{n}$

Although $\mathbb{C} \simeq \mathbb{C}_p$ as fields. (as they have the <u>same transcendental degree</u> over $\overline{\mathbb{Q}}$)

$\Uparrow$ the same cardinality

they carry totally different topology.

Euclidean top. on $\mathbb{C}$ (connected)

Non-arch. top. on $\mathbb{C}_p$ (totally disconnected)

§1.3 <u>Hodge-Tate decomposition</u>

Given $X / \mathbb{C}_p$ projective smooth variety

$$\begin{array}{ccc} X_{\mathbb{C}} & \simeq & X \\ | & & | \\ \mathbb{C} & \simeq & \mathbb{C}_p \end{array}$$

Euclidean top. on $X_{\mathbb{C}}$ as manifold

$\neq$ étale / Zariski topology on X as scheme

We have comparison isomorphism (for any $\mathbb{C} \simeq \mathbb{C}_p$)

$$H^n_{sing}(X_{\mathbb{C}}, \mathbb{C}) \simeq H^n_{\acute{e}t}(X, \mathbb{C}_p) := \left(\varprojlim_m H^n_{\acute{e}t}(X, \mathbb{Z}/p^m) \right) \otimes_{\mathbb{Z}_p} \mathbb{C}_p$$

(Artin, SGA 4, XI)

$\Rightarrow$ the terms in Hodge decomposition come from algebraic geometry.

$$H^n_{\acute{e}t}(X, \mathbb{C}_p) \simeq \bigoplus_{i+j=n} H^j(X, \Omega^i_{X/\mathbb{C}_p})$$

but this isom. depends on the arbitrary choice of $\mathbb{C} \simeq \mathbb{C}_p$.

Q. Is there a purely algebraic proof / a canonical isom?

How is the valuation ring str. $\mathbb{Q}_p \supset \mathbb{Z}_p$ involved here?

<u>History</u>. Tate 1967 : abelian variety $/K \mid \mathbb{Q}_p$ (finite) w/ good reduction

Faltings 1988 : proper smooth var. $/K$ (almost purity + Galois coh.)

Scholze 2013 : proper smooth var. $/\mathbb{C}_p$ (perfectoid theory)

How to understand p-adic coh ?

The key is $\mathbb{Q}_p \supset \mathbb{Z}_p$ and the <u>ramification</u> above it.

Before we try to understand ramification above X,

<u>§ 1.4. ramification of $\overline{\mathbb{Q}_p}$ over $\mathbb{Q}_p$</u>

<u>Example</u> (cyclotomic ext'n)

$$\begin{tikzcd}
\mathbb{Q}_p(\zeta_{p^\infty}) \arrow[r, dash] \arrow[d, dash, bend right=60, "\mathbb{Z}_p^\times"'] \arrow[d, dash, bend right=30, "\mathbb{Z}_p"'] & \mathbb{Z}_p[\zeta_{p^\infty}] \arrow[d, dash] \\
\mathbb{Q}_p(\zeta_{p^2}) \arrow[r, dash] \arrow[d, dash, "p"'] & \mathbb{Z}_p[\zeta_{p^2}] \arrow[d, dash, "\text{tot. ramified}"] \\
\mathbb{Q}_p(\zeta_p) \arrow[r, dash] \arrow[d, dash, "p-1"'] & \mathbb{Z}_p[\zeta_p] \arrow[d, dash, "\text{tot. ramified}"] \\
\mathbb{Q}_p \arrow[r, dash] & \mathbb{Z}_p
\end{tikzcd}$$

	uniformizer π	$v_p(\pi)$
$\vdots$	$\vdots$	$\vdots$
ζ_{p^2} = primitive p^2-th root of 1	$\zeta_{p^2}-1$	$\frac{1}{p(p-1)}$
ζ_p = primitive p-th root of 1	ζ_p-1	$\frac{1}{p-1}$
	p	1

$v_p(D_{K_\cdot/K})$

$\vdots$

$2-\frac{1}{p-1}$

$1-\frac{1}{p-1}$

0

How to understand ramification of $L \,|\, \mathbb{Q}_p(\zeta_{p^\infty}) = K$?

$$\begin{tikzcd}
L \arrow[r, dash] & \cdots \arrow[r, dash] & L_2 \arrow[r, dash] & L_1 \arrow[r, dash] & L_0 \arrow[dl, dash, "\text{finite Galois}"'] \\
K_\infty \arrow[u, dash, "\text{finite Galois}"] \arrow[r, dash] \arrow[rrr, dash, bend right, "\text{totally ramified } \mathbb{Z}_p\text{-ext.}"'] & \cdots \arrow[r, dash] & K_2 \arrow[r, dash] & K_1 \arrow[r, dash] & K_0 = \mathbb{Q}_p(\zeta_{p^r}), \quad r \gg 0
\end{tikzcd}$$

1967

<u>Th. (Tate)</u> (1) $v_p(D_{K_n|K_0}) = n + c + p^{-n} a_n \quad , \forall n$

where c is constant, $\{a_n\}$ bounded sequence

(2) $v_p(D_{L_n|K_n}) \to 0 \quad (n \to \infty)$

$\Rightarrow \left(D_{L|K_\infty}^{a} \subset \mathcal{O}_L \right)$

i.e. L is "almost unramified" over K_∞.

ramification of $\overline{\mathbb{Q}}_p$ over $\mathbb{Q}_p$ $\overset{\text{almost}}{\approx}$ $\mathbb{Q}_p(\zeta_{p^\infty}) \,|\, \mathbb{Q}_p$.

<u>Upshot</u>,

Tate

$$\begin{tikzcd}
\bar{K} \arrow[d, dash, "\text{almost unramified}"] \\
K_\infty \arrow[d, dash, "\text{regularly ramified}"] \\
K
\end{tikzcd}
\quad \overset{\text{Fontaine}}{\rightsquigarrow} \quad
\begin{tikzcd}
\bar{R} \arrow[d, dash, "\text{almost unramified (almost purity)}"] \\
R_\infty \text{ - pre-perfectoid} \arrow[d, dash, "\text{regularly ramified (Galois coh. computation)}"] \\
R \text{ smooth} / \mathbb{Q}_p
\end{tikzcd}$$

(local class field theory)

§2. Definition of perfectoids

§2.1 Review of deformation theory

Thickening. $\operatorname{Spec} R \hookleftarrow \operatorname{Spec} R_0$, $R_0 = R/I$

$I^2 = 0$

(e.g. $\operatorname{Spec} \mathbb{Z}_p$ ..... $\hookleftarrow \operatorname{Spec} \mathbb{Z}/p^3\mathbb{Z} \hookleftarrow \operatorname{Spec} \mathbb{Z}/p^2\mathbb{Z} \hookleftarrow \operatorname{Spec} \mathbb{F}_p$)

Lifting problem.

$$\begin{tikzcd} \operatorname{Spec} A \arrow[d, dashed, "\text{flat}"'] \arrow[dr, phantom, "?"] & \operatorname{Spec} A_0 \arrow[l, dashed] \arrow[d, "\text{flat}"] \\ \operatorname{Spec} R & \operatorname{Spec} R_0 \arrow[l, hook'] \end{tikzcd}$$

Example. $A_0 = R_0[T]$ there is an obvious lift $A = R[T]$.

$$\begin{tikzcd} A \ni T \arrow[r, mapsto] & T \in A_0 \\ A = R[T] \arrow[r] & A_0 = R_0[T] \arrow[d] \\ R \arrow[u] \arrow[r] & R_0 \end{tikzcd}$$

Any other lifting ?

Any automorphism ?

$$\begin{tikzcd} \sigma \circlearrowright R[T] \arrow[r, two heads] & R_0[T] \circlearrowleft \mathrm{id} \\ R \arrow[u] \arrow[r, two heads] & R_0 = R/I \arrow[u] \end{tikzcd}$$

$T \mapsto T + i$, $i \in \ker(R \to R_0)$

$\simeq I \cdot A$

For general A_0, resolve A_0 by polynomial algebras

$$\cdots \rightrightarrows P_2 \rightrightarrows P_1 = R_0[R_0[A_0]] \rightrightarrows P_0 = R_0[A_0] \to A_0 \to 0$$

$T_a \mapsto a$

$T_{rT_a} \mapsto r \cdot T_a$, $T_{rT_a} \mapsto T_{r \cdot a}$

$R_0[T_a : a \in A_0]$

R_0 - algebra

Cotangent complex

$$\mathbb{L}_{A_0/R_0} := \left(\cdots \to \Omega_{P_1} \otimes_{P_1} A_0 \to \Omega_{P_0} \otimes A_0 \right)$$

Example. $A_0 = R_0[T]$,

$\Omega_{R_0[T]/R_0} = R_0[T]\, dT$

$\mathbb{L}_{A_0/R_0} = \Omega_{A_0/R_0} \simeq A_0\, dT$ is free A_0-mod. of rank 1.

Th. 1 (Illusie 1971) Complex cotangent et déformations

(1) $\exists\ \omega \in \mathrm{Ext}^2(\mathbb{L}_{A_0/R_0}, A_0 \otimes_{R_0} I)$, s.t.

$\omega = 0 \iff$ exists a flat lifting A

(2) $\{$isom. classes of liftings $A\}$ is torsor under $\mathrm{Ext}^1(\mathbb{L}_{A_0/R_0}, A_0 \otimes_{R_0} I)$

(3) $\{$automorphisms of lifting $A\} = \mathrm{Ext}^0(\mathbb{L}_{A_0/R_0}, A_0 \otimes_{R_0} I)$.

$$\begin{tikzcd}
\mathrm{Spec}\, A' \arrow[d, dashed, "f\ ?"] & \mathrm{Spec}\, A_0' \arrow[l] \arrow[d, "f_0"] \\
\mathrm{Spec}\, A \arrow[d] & \mathrm{Spec}\, A_0 \arrow[l] \arrow[d] \\
\mathrm{Spec}\, R & \mathrm{Spec}\, R_0 \arrow[l]
\end{tikzcd}$$

Th. 2 (Illusie)

(1) $\exists\ \omega \in \mathrm{Ext}^1(\mathbb{L}_{A_0/R_0}, A_0' \otimes_{R_0} I)$ s.t.

$\omega = 0 \iff \exists$ lifting f

(2) $\{$liftings $f\}$ is a torsor under $\mathrm{Ext}^0(\mathbb{L}_{A_0/R_0}, A_0' \otimes_{R_0} I)$

$\mathbb{L}_{A/R} = 0 \iff$ formally étale

$\mathbb{L}_{A/R}$ is deg 0 projective mod $\iff$ formally smooth

$H^0(\mathbb{L}_{A/R}) = \Omega^1_{A/R} = 0 \iff$ formally unramified.

§2. Definition of perfectoids

§2.2 Universal p-deformation: Witt rings

We fix a perfect $\mathbb{F}_p$-algebra R : $\mathrm{Frob} : R \xrightarrow{\sim} R$
$x \mapsto x^p$

Lemma. $\mathbb{L}_{R/\mathbb{F}_p} = 0$.

Proof. Standard resolution

$$\begin{tikzcd}
\cdots \arrow[r] & P_1 \arrow[r] \arrow[d, "\mathrm{Frob}"] & P_0 \arrow[r] \arrow[d, "\mathrm{Frob}"] & R \arrow[r] \arrow[d, "\wr\ \mathrm{Frob}"] & 0 \\
\cdots \arrow[r] & P_1 \arrow[r] & P_0 \arrow[r] & R \arrow[r] & 0
\end{tikzcd}$$

$$\begin{tikzcd}
\mathbb{L}_{R/\mathbb{F}_p} = (\cdots \arrow[r] & \Omega_{P_1/\mathbb{F}_p} \otimes_{P_1} R \arrow[r] \arrow[d, "\mathrm{Frob}\ =\ 0"] & \Omega_{P_0/\mathbb{F}_p} \otimes_{P_0} R) \arrow[d, "\mathrm{Frob}\ =\ 0"] & dx \arrow[d, mapsto] \\
\mathbb{L}_{R/\mathbb{F}_p} = (\cdots \arrow[r] & \Omega_{P_1/\mathbb{F}_p} \otimes_{P_1} R \arrow[r] & \Omega_{P_0/\mathbb{F}_p} \otimes_{P_0} R) & d(x^p) = px^{p-1}dx = 0
\end{tikzcd}$$

$\Rightarrow \mathbb{L}_{R/\mathbb{F}_p} = 0$.

Lecture 2 §2. Definition of perfectoid

§2.2. Universal p-deformation: Witt rings

We fix a perfect $\mathbb{F}_p$-algebra R

Lemma. $\mathbb{L}_{R/\mathbb{F}_p} = 0$ in $D(R)$.

there are unique flat liftings.

$$\begin{tikzcd} \cdots \arrow[r] & R_3 \arrow[r] & R_2 \arrow[r] & R_1 = R \\ \cdots \arrow[r] & \mathbb{Z}/p^3\mathbb{Z} \arrow[r] \arrow[u] & \mathbb{Z}/p^2\mathbb{Z} \arrow[r] \arrow[u] & \mathbb{F}_p \arrow[u] \end{tikzcd}$$

<u>Def'n.</u> We denote by $W(R) = \varprojlim R_n$. (also called the Witt ring of R)

the unique p-adically complete and flat $\mathbb{Z}_p$-algebra s.t. (i.e. p-torsion free)

$$W(R)/pW(R) = R.$$

<u>Rmk.</u> By deformation theory, any morphism of perfect $\mathbb{F}_p$-algebras $R \to R'$ lifts uniquely to $W(R) \to W(R')$. i.e. there is an equiv. of cat.

$$\left\{ \begin{array}{l} p\text{-complete flat } \mathbb{Z}_p\text{-alg } A \\ \text{s.t. } A/pA \text{ perfect} \end{array} \right\} \xrightarrow{\sim} \left\{ \text{perfect } \mathbb{F}_p\text{-alg} \right\}$$

$$A \longmapsto A/pA$$

$$W(R) \longleftarrow\!\shortmid R$$

<u>Prop.</u> (Teichmüller lifting) There is a unique multiplicative section

$[\ ]: R \to W(R)$ of $W(R) \twoheadrightarrow R$.

<u>Proof.</u> <u>Existence:</u> $W(R) \to W(R)/p^n W(R) \to W(R)/pW(R) = R$ perfect

$$(y_n + p\,\xi)^{p^n} \longmapsto x$$

does not depend on ξ

$$y_n + p\,\xi \longmapsto x^{1/p^n}$$

$\xi \in W(R)$

$$(y_n + p\,\xi)^{p^n} \equiv y_n^{p^n} \mod p^n W(R)$$

$W(R) = \varprojlim W(R)/p^n W(R)$

$[x] = (\cdots, y_2^{p^2}, y_1^{p}, y_0 = x)$

It's easy to see that $[xx'] = [x][x']$.

<u>Uniqueness</u> $\forall x \in R$, $[x]' \equiv [x] \mod pW(R)$

$$[x]' = [x] + p\xi$$

$$\Rightarrow [x^{p^n}]' = ([x] + p\xi)^{p^n} \equiv [x^{p^n}] \mod p^n W(R)$$

Since R perfect, any elt is of the form x^{p^n}

$$\Rightarrow \forall x \in R, \quad [x]' \equiv [x] \mod p^n W(R)$$

Let $n \to \infty$, $[x]' = [x]$ in $W(R)$.

<u>Cor.</u> (Teichmüller expansion)

$\forall x \in W(R)$, there is a unique series

$$x = [x_0] + p[x_1] + p^2[x_2] + \cdots$$

In particular, $x \in W(R)^\times$ $\Leftrightarrow$ $x_0 \in R^\times$.

(because $W(R)$ is p-adically complete)

<u>Proof.</u> Let x_0 be the image of $x \in W(R)$ in R.

$$\Rightarrow x = [x_0] + px', \quad x' \in W(R)$$

(where x' is unique as $W(R)$ is p-torsion free)

$$x' = [x_1] + px''$$

$$x'' = [x_2] + px''' \quad \cdots$$

<u>Rmk.</u> $W(R) \xrightarrow{\sim} \prod_{n=0}^{\infty} R$

$$[x_0] + p[x_1] + p^2[x_2] + \cdots \longmapsto (x_0, x_1^p, x_2^{p^2}, \ldots)$$

The latter is the usual ring of Witt vectors.

§ 2.3 Universal ξ-deformation: perfect prisms

Upshot:

$$\begin{tikzcd} W(R) \arrow[r, two heads, "\text{univ. } p\text{-def}"] \arrow[d, two heads, "\text{univ } \boxed{\xi}\text{-def}"] & R \quad \text{perfect} \\ S \quad \text{perfectoid ring} / \mathbb{Z}_p & \end{tikzcd}$$

Def'n A <u>perfect prism</u> is a pair (A, I) of ring A w/ ideal I s.t.

(1) A is a p-complete flat $\mathbb{Z}_p$-alg w/ $R = A/pA$ perfect (i.e. $A = W(R)$)

(2) $I = (\xi)$ for $\xi = [\xi_0] + p[\xi_1] + p^2[\xi_2] + \cdots$

s.t. R is ξ_0-complete and $\underline{\xi_1 \in R^\times}$. ($\Rightarrow$ ξ is non-zero divisor in A)

Prop. $\{\text{perfect prisms } (A, I)\} \longrightarrow \{\text{rings}\}$ fully faithful

$$(A, I) \longmapsto A/I$$

Proof. For $S = A/I$, need to reconstruct (A, I)

$$\begin{tikzcd} A = W(R) \arrow[r, two heads] \arrow[d, "I = (\xi)"'] & A/p = R \arrow[d] \\ S = A/I \arrow[r, two heads] & S/p = R/\xi_0 \end{tikzcd}$$

need to reconstruct

Lemma. For d-complete ring R, $\varprojlim_{\text{Frob}} R \xrightarrow{\sim} \varprojlim_{\text{Frob}} R/d$.

$y_n \equiv y_n' \mod dR$

$$\begin{array}{ccccc} (\cdots, & x_2, & x_1, & x_0) & \\ & \uparrow & \uparrow & \uparrow & \\ \cdots & y_2 & y_1 & y_0 & \\ & y_2' & y_1' & y_0' & \end{array}$$

$$\Rightarrow \quad y_n^{p^m} \equiv y_n'^{\,p^m} \quad (\mathrm{mod}\ d^{p^m} R)$$

$$\Rightarrow \quad \lim_{m\to\infty} y_{n+m}^{p^m} \in R$$

$$\left(\cdots, \lim_{n\to\infty} y_{n+1}^{p^n}, \lim_{n\to\infty} y_n^{p^n} \right) \longleftarrow \left(\cdots, x_2, x_1, x_0 \right)$$

$$\begin{array}{ccc} & & \uparrow \\ & & y_2 \quad y_1 \quad y_0 \end{array}$$

$$\begin{tikzcd} W(S^\flat) \arrow[r] \arrow[d, "\theta"'] & S^\flat = \varprojlim_{\mathrm{Frob}} S/p \arrow[d] \\ S \arrow[r] & S/p \end{tikzcd} \qquad \text{perfect } \mathbb{F}_p\text{-alg}$$

$$(A, I) = \left(W(S^\flat), \ker(\theta) \right)$$

<u>Def'n</u> A <u>perfectoid ring</u> is a ring S s.t. $S \simeq A/I$ for some perfect prism (A, I).

<u>§2.4. Tilting correspondence of perfectoids</u>

<u>Th.</u> Given a perfect prism (A, I),

$$\begin{tikzcd} A = W(R) \arrow[r, two heads] \arrow[d, two heads] & A/p = R \arrow[d, two heads] \\ S = A/\xi \arrow[r, two heads] & S/p = R/\xi_0 \end{tikzcd}$$

$$\left\{ \text{perfect prisms } (A', IA')/(A, I) \right\} \xrightarrow[\sim]{①} \left\{ \xi_0\text{-complete perfect } R\text{-alg } R' \right\}$$

$$\searrow \left\{ \text{perfectoid } S\text{-alg } S' \right\}$$

<u>Proof.</u> ① by deformation over $A \to R$ & $\mathbb{L}_{R'/R} = 0$.

$$\begin{tikzcd} \left\{ \text{perfect prisms } (A', IA')/(A, I) \right\} \arrow[r, "①", "\sim"'] \arrow[d, "④"', "\wr"] & \left\{ \xi_0\text{-complete perfect } R\text{-alg } R' \right\} \arrow[d, "②"', "\wr"] \\ \left\{ \text{perfectoid } S\text{-alg } S' \right\} \arrow[r, "③", "\sim"'] & \left\{ \text{relative perfect } (S/p) = (R/\xi_0)\text{-alg } T \right\} \end{tikzcd}$$

$$T \longrightarrow T \otimes_{S/p} S/p \xrightarrow{\sim} T$$

$$T \xrightarrow{\mathrm{Frob}} S/p$$

② by deformation over $R \to R/\xi_0$ & $\mathbb{L}_{T/(R/\xi_0)} = 0$

③ by deformation over $S \twoheadrightarrow S/p$ & $\mathbb{L}_{T/(S/p)} = 0$

$\Rightarrow$ ④ also equiv.

<u>Rmk</u> We write $R' = S'^{\flat}$ and $S' = R'^{\#}$.

§ 2.5 Properties of perfectoids

<u>Def'n</u> Let S be a perfectoid ring, a <u>strict uniformizer</u> ϖ is the image of $[\xi_0] \in W(R)$

$$\begin{tikzcd} {[\xi_0]} \arrow[d, mapsto] & W(R) \arrow[r] \arrow[d, "\theta"'] & R \ni \xi_0 \arrow[d] \\ \varpi & S \arrow[r] & S/p = R/\xi_0 \end{tikzcd}$$

<u>Lem.</u> (p^∞-roots). $\underline{\underline{\varpi \in pS^\times}}$ and admits a compatible p^∞-roots. $\varpi^{1/p^n} = \theta([\xi_0^{1/p^n}])$

$\xi = [\xi_0] + p[\xi_1] + \cdots$

<u>proof</u> Because $\xi_1 \in R^\times$, $\xi = [\xi_0] + p\xi'$, $\xi' \in W(R)^\times$

mod $\xi W(R)$ $\Rightarrow$ $0 = \varpi + p\theta(\xi')$ in S

<u>Lem.</u> (Frob. isom.) $S/\varpi^{1/p}S \xrightarrow[\sim]{\text{Frob}} S/\varpi S$.

<u>Proof.</u> $S/pS = R/\xi_0 R$ & R perfect

$\varpi \longmapsto \xi_0$

$\varpi^{1/p} \longmapsto \xi_0^{1/p}$

<u>Lem</u> (Non-zero div) R ξ_0-complete perfect $\mathbb{F}_p$-alg. $\xi = [\xi_0] + p\xi'$, $\xi' \in W(R)^\times$ (distinguished)

Then ξ is a nonzero divisor in $W(R)$.

Proof. Suppose $0 = \xi \cdot x = \left([\xi_0] + p\xi'\right) \cdot x$

n odd $\Rightarrow \left([\xi_0]^n + (p\xi')^n\right) \cdot x = 0$

$$\left([\xi_0^n] + p^n \xi'^n\right) \cdot x$$

$$\Rightarrow p^n x \in [\xi_0^n] \cdot W(R)$$

$$x = [x_0] + p[x_1] + p^2[x_2] + \cdots$$

$$p^n x = p^n[x_0] + p^{n+1}[x_1] + p^{n+2}[x_2] + \cdots$$

$$= [\xi_0^n]\left([y_0] + p[y_1] + p^2[y_2] + \cdots\right)$$

$$= [\xi_0^n y_0] + p[\xi_0^n y_1] + p^2[\xi_0^n y_2] + \cdots$$

$\Rightarrow \forall i \geq 0, \quad x_i \in \xi_0^n R. \quad \Rightarrow x_i \in \bigcap_{n \text{ odd}} \xi_0^n R = 0 \Rightarrow x = 0.$

Lemma (completeness) $W(R)$ is (p, ξ)-complete, S is p-complete.

Proof. $0 \to W(R)/p^{n-1} \to W(R)/p^n \to \underbrace{W(R)/pW(R) = R}_{\xi_0\text{-complete perfect } \mathbb{F}_p\text{-alg}} \to 0$

$\to W(R)/p^n$ is ξ_0-complete

$$\Rightarrow W(R) = \varprojlim_n W(R)/p^n = \varprojlim_{n,m} W(R)/(p^n, \xi^m) = \varprojlim_n W(R)/(p^n, \xi^n)$$

$$= \varprojlim_n W(R)/(p, \xi)^n$$

$\Rightarrow W(R)$ is (p, ξ)-complete, $\quad S = W(R)/\xi$ is p-complete.

Lem (Almost torsion-free) $S[\varpi^{\infty}] = S[\varpi^{1/p^{\infty}}]$, i.e. p^{∞}-torsion of S is almost zero (killed by ϖ^{1/p^n})

Proof. p, ξ are non-zero divisors on A

$$\begin{array}{ccc} (A/\xi)[p] & \simeq & (A/p)[\xi] \\ \| & & \| \\ S[p] & & R[\xi_0] \\ \| & & \\ S[\varpi] & & \end{array}$$

$$\begin{array}{ccc} S[\varpi] & = & R[\xi_0] \\ \cup & & \cup \ \| \longleftarrow \text{as } R\text{- perfect} \\ S[\varpi^{1/p^n}] & = & R[\xi_0^{1/p^n}] \end{array}$$

R perfect, $R[\xi_0^{1/p^{\infty}}] = R[\xi_0] = R[\xi_0^{p^{\infty}}]$

$\Rightarrow \quad S[\varpi^{1/p^{\infty}}] = S[\varpi] = S[\varpi^{p^{\infty}}]$.

Prop. (Perfectoid criterion) S p-torsion free ring. is perfectoid if and only if

① S p-complete

② $\exists\ \pi \in S$ s.t. $\pi^p \in pS^{\times}$.

③ $S/\pi S \xrightarrow[\sim]{\text{Frob}} S/pS$.

Proof.

$$\begin{array}{ccc} W(S^b) & \longrightarrow & S^b = \varprojlim_{\text{Frob}} S/p \\ \theta \downarrow & & \downarrow \text{surj. by ③} \\ S & \longrightarrow & S/p \end{array}$$

$\exists\ \pi^b = (\cdots, \pi_1, \pi, 0) \in S^b = \varprojlim_{\text{Frob}} S/p$

② $\Rightarrow S^b/\pi^b S^b \xrightarrow[\sim]{} S/pS$

Lecture 3 §3. Cohomological descent of perfectoids

Last time.

$$\begin{tikzcd}
A = W(R) \arrow[r, "\text{univ. } p\text{-def.}"] \arrow[d, "\text{univ. } \xi\text{-def.}"] & \overset{A/p}{\overset{\shortparallel}{R}} \arrow[d] \\
A/\xi = S \arrow[r, two heads] & S/p \simeq R/\xi_0
\end{tikzcd}$$

ξ distinguished, $\xi = [\xi_0] + p\underbrace{\xi'}_{\text{unit}}$ — R perfect $\mathbb{F}_p$-algebra, ξ_0-complete

$A/\xi = S$ perfectoid

Today. transfer coh. desc. of perfect $\mathbb{F}_p$-alg to that of perfectoids.

Recall. Zariski descent

$X = \operatorname{Spec} A$, $\quad A \xrightarrow{\sim} R\Gamma_{Zar}(X, \mathcal{O}_X)$

equivalently, $\quad A \longrightarrow \prod_{i=1}^{n} A_{f_i} \longrightarrow \prod_{1 \le i,j \le n} A_{f_i f_j} \longrightarrow \cdots$ exact

- faithfully flat descent (Grothendieck)

$$A \xrightarrow[\text{f. flat}]{} B \longrightarrow B \otimes_A B \longrightarrow B \otimes_A B \otimes_A B \longrightarrow \cdots \quad \text{exact}$$

i.e. $\quad A \xrightarrow{\sim} R\Gamma_{fppf}(X, \mathcal{O}_X)$.

History. - h-descent (Gabber)

A perfect $\mathbb{F}_p$-alg.

$$A \xrightarrow[\text{h-cover}]{} B \longrightarrow B \otimes_A B \longrightarrow \cdots \quad \text{exact}$$

(a refinement of fppf and proper surj.)

- v- descent & arc-descent (Bhatt-Mathew 2018)

- arc descent for perfectoid rings (Bhatt-Scholze, 2019)

§3.1 h, v and arc topologies

Def'n A <u>valuation ring</u> is a local domain V s.t. $x \mid y$ or $y \mid x$, $\forall x, y \in V$.

An <u>ext'n of valuation rings</u> is an <u>injective local</u> homomorphism $V \to W$ of valuation rings.

$\Updownarrow$

faithfully flat

Prop (1) $K = \mathrm{Frac}(V)$, $K^\times / V^\times$ is totally ordered w.r.t. divisibility.

(Rmk. DVR V, $K^\times / V^\times \simeq \mathbb{Z}$)

(2) $K^\times \xrightarrow{v} K^\times / V^\times$ is a valuation map $\left(v(x+y) \geq \min(v(x), v(y)) \right)$

(3) The ideals of V form a chain

(4) V of dim ≤ 1 $\iff$ $K^\times / V^\times \subset \mathbb{R}$.

(5) V DVR $\iff$ $K^\times / V^\times \simeq \mathbb{Z}$.

Def. $A \to B$ a ring homomorphism is

① a <u>v-covering</u> if

$$\begin{tikzcd} B \arrow[r, dashed, "\exists"] & W \\ A \arrow[u] \arrow[r] & V \arrow[u] \end{tikzcd}$$

ext'n of val. ring.

valuation ring

② an <u>arc-covering</u> (d-complete) if the above condition holds for V of dim ≤ 1. (d-complete)

③ an <u>h-covering</u> if v-covering, locally of finite presentation.

$\{h\text{-cover}\} \subset \{v\text{-cover}\} \subset \{arc\text{-cover}\} \subset \{d\text{-complete arc cover}\}$

$\|$

$\{$faithfully flat cov. + proper surj.$\}$

$\cup$

$\{$fppf cover$\}$

$\cup$

$\{$étale cover$\}$

$\cup$

$\{$Zar. cover$\}$

<u>Example</u>. A ring

$\forall\, x \in \operatorname{Spec} A$, $k(x)$ the residue field $= A_x/m_x$

$k(x) \supset V$ val. ring s.t. $k(x) = \operatorname{Frac} V$.

$\mathcal{V} = \{V\}$ forms a <u>set</u>.

$$\begin{tikzcd} A \arrow[r] \arrow[dr] & \prod_{\mathcal{V}} V \arrow[d, dashed] \\ & V' \end{tikzcd}$$

is a v-cover.

$\overline{k(x)}$ alg. closure $\supset \overline{V}$ ext. of V

$$\begin{tikzcd} A \arrow[r] \arrow[d] & \prod_{\mathcal{V}} \overline{V} \arrow[d, dashed] \\ V' \arrow[r] & \overline{V'} \end{tikzcd}$$

is a v-cover.

§3.2 arc-descent for perfect $\mathbb{F}_p$-alg.

A $\mathbb{F}_p$-algebra

$$A_{perf} = \operatorname*{colim}_{Frob} A = \operatorname{colim}\left(A \xrightarrow{Frob} A \xrightarrow{Frob} A \xrightarrow{Frob} \cdots\right)$$ the initial perfect A-algebra.

<u>Th.</u> (Gabber) For any h-covering $A \to B$,

$$0 \to A_{perf} \to B_{perf} \to (B \otimes_A B)_{perf} \to \cdots \quad \text{exact.}$$

In particular, if A, B perfect, $0 \to A \to B \to B \otimes_A B \to \cdots$ exact.

<u>Example 1</u>.

$$\begin{tikzcd} E = \operatorname{Spec} A/fA+J \arrow[r] \arrow[d, hook] & \operatorname{Spec} A/J = X' \arrow[d, hook] \\ Z = \operatorname{Spec} A/fA \arrow[r, hook] & \operatorname{Spec} A = X \end{tikzcd}$$

where $f^n J = 0$, i.e. $A = A/f^n J$

X is covered in h-topology by the two closed subschemes Z and X', with scheme theoretic intersection E.

We want

$$0 \to A_{perf} \to (A/fA)_{perf} \oplus (A/J)_{perf} \to (A/fA+J)_{perf} \to 0$$

$$a \longmapsto (x \in A/fA,\ y \in A/J) \longmapsto 0 \quad \text{exact.}$$

i.e. $Frob^n(x) \equiv Frob^n(y) \mod fA+J$

$Frob^n(x) - Frob^n(y) = fz + g$, $z \in A$, $g \in J$

$$\underbrace{Frob^n(x) - fz}_{a} = Frob^n(y) - g$$

Example 2

$$\begin{tikzcd}
E = \mathrm{Proj}\left(A/(f_1 f_2)[T_1,T_2]\right) \arrow[r] \arrow[d] & \mathrm{Proj}\left(A[T_1,T_2]/(T_1 f_2 - T_2 f_1)\right) = X' \arrow[d] & \mathbb{P}^1_X \\
Z = \mathrm{Spec}\, A/(f_1 f_2) \arrow[r, hook] & \mathrm{Spec}\, A = X &
\end{tikzcd}$$

$\mathbb{P}^1_Z \simeq$ $\qquad$ $T_1 f_2 = T_2 f_1$

① $H^0(E, \mathcal{O}_E) \xrightarrow{\sim} R\Gamma(E, \mathcal{O}_E)$, $\quad H^0(X', \mathcal{O}_{X'}) \xrightarrow{\sim} R\Gamma(X', \mathcal{O}_{X'})$

$$\begin{tikzcd}
H^0(E, \mathcal{O}_E) \arrow[d, "\wr"] & H^0(X', \mathcal{O}_{X'}) \arrow[d, two heads] \\
H^0(Z, \mathcal{O}_Z) = A/(f_1 f_2) & H^0(X, \mathcal{O}_X) = A
\end{tikzcd}$$

with kernel square-zero

$\Downarrow$

② $$\begin{tikzcd}
0 \arrow[r] & H^0(X, \mathcal{O}_X)_{perf} \arrow[r] \arrow[rr, bend right, "\sim"'] & H^0(Z, \mathcal{O}_Z)_{perf} \oplus H^0(X', \mathcal{O}_{X'})_{perf} \arrow[r] \arrow[rr, bend left, "\sim"] & H^0(E, \mathcal{O}_E)_{perf} \arrow[r] & 0
\end{tikzcd}$$

$$\ker\left(A \longrightarrow A[X]/(f_2 X - f_1) \oplus A[Y]/(f_2 - f_1 Y)\right)^2 = 0$$

$a = (f_2 X - f_1)(a_0 + a_1 X + \cdots + a_n X^n)$

$b = (f_2 - f_1 Y)(b_0 + b_1 Y + \cdots + b_m Y^m)$

$\Rightarrow a = -f_1 a_0$, $\quad f_2 a_0 = f_1 a_1$, $\quad f_2 a_1 = f_1 a_2, \cdots, f_2 a_{n-1} = f_1 a_n$, $f_2 a_n = 0$

$b = f_2 b_0$, $\quad f_1 b_0 = f_1 f_2$, $\quad f_1 b_1 = f_2 b_2, \cdots, f_1 b_{m-1} = f_2 b_m$, $\quad f_1 b_m = 0$

$\Rightarrow -ab = f_1 f_2 a_0 b_0 = f_1 f_2 a_1 b_1 = f_1 f_2 a_2 b_2 = \cdots = f_1 f_2 a_N b_N = 0$.

<u>Th. (Gabber)</u> $\forall$ h-covering $A \to B$ of $\mathbb{F}_p$-algebras,

$$0 \longrightarrow A_{perf} \longrightarrow B_{perf} \longrightarrow (B \otimes_A B)_{perf} \longrightarrow \cdots \quad \text{exact}$$

(or equivalently, $\forall$ $\mathbb{F}_p$- scheme X,

$$H^q_h(X, \mathcal{O}_{X\,perf}) \simeq H^q_{Zar}(X, \mathcal{O}_X)_{perf} \quad).$$

<u>Idea</u>. By abstract formalism (decomposing h- cover by almost blowups), we reduce to Examples 1 & 2.

<u>Cor</u> (Bhatt- Mathew 2018) For any arc-covering $A \to B$ of $\mathbb{F}_p$-algebras,

$$0 \to A_{perf} \to B_{perf} \to (B \otimes_A B)_{perf} \to \cdots \quad \text{exact.}$$

<u>Idea</u> h- descent $\Rightarrow$ v- descent by a limit process.

v- descent $\Rightarrow$ arc- descent by dividing a valuation ring into arcs.

<u>Rmk</u>. $X = \operatorname{Spec} R$, R $\mathbb{F}_p$- alg

$\mathcal{O}$: arc sheafification of $[X \mapsto R]$.

$\forall X$ admits an arc cover by perfect schemes.

$$\Rightarrow \quad \mathcal{O} = \operatorname*{colim}_{Frob} \mathcal{O} = \mathcal{O}_{perf}.$$

$$\boxed{R\Gamma_{arc}(X, \mathcal{O}) = R_{perf}.}$$

$$R\Gamma_{arc}(X, \mathcal{O}) = R\Gamma_{Zar}(X, \mathcal{O})_{perf} \quad \text{in general.}$$

<u>§ 3.3 Tilting correspondence of perfectoid valuation rings</u>

<u>Lemma</u>. Let S be a perfectoid ring, $S^\flat = \varprojlim_{Frob} S/pS$.

Then S is valuation ring $\Leftrightarrow$ $S^\flat$ is a valuation ring.

Proof $\Rightarrow$ S p-complete $\xrightarrow{\text{Lem.}}$ $\varprojlim_{\text{Frob}} S \xrightarrow{\sim} \varprojlim_{\text{Frob}} S/pS = S^\flat$ multiplicative bijection.

$$\left(\ldots, \lim_n y_{n+2}^{p^n}, \lim_n y_{n+1}^{p^n}, \lim_n y_n^{p^n}\right) \qquad (\ldots, x_2, x_1, x_0)$$

$$(\ldots, y_2, y_1, y_0)$$

S domain $\Rightarrow$ $S^\flat$ domain.

$\forall x, y$, $x \mid y$ or $y \mid x$ for S $\Rightarrow$ for $S^\flat$

$S/pS = S^\flat / \xi_0 S^\flat$, S local $\to$ $S^\flat$ local.

Prop. Let S be perfectoid w/ tilt $S^\flat$. $\varpi = [\xi_0]$ strict pseudo uniformizer

$\xi_0 = \varpi^\flat = (\ldots, \varpi^{1/p^2}, \varpi^{1/p}, \varpi)$

$$\{\text{perfectoid valuation ring } /S\} \xrightarrow{\sim} \{\varpi^\flat\text{-complete perfect val. ring } /S^\flat\}$$

$$V \longmapsto V^\flat$$

Proof. Follows directly from tilting correspondence

Prop. $S \to S'$ morphism of perfectoid rings.

$S^\flat \to S'^\flat$ morphism of $\varpi^\flat$-flat complete perfect $\mathbb{F}_p$-alg

Then $S \to S'$ is a ϖ-complete arc cover

$\Leftrightarrow$ $S^\flat \to S'^\flat$ is a $\varpi^\flat$-complete arc cover.

Proof. $\Leftarrow$

$$\begin{tikzcd} S' \\ S \arrow[u] \arrow[r] & V \end{tikzcd}$$

ϖ-complete of $\dim \leq 1$

Frac V alg. closed.

Case 1 $\varpi = 0$ in V, i.e.

$$\begin{tikzcd}
S' \arrow[r] & S'/\varpi \arrow[r, dashed] & \exists W & S'^b/\varpi^b \arrow[l] \arrow[ll, bend right, "\sim"'] & S'^b \arrow[l] \\
S \arrow[u] \arrow[r] & S/\varpi \arrow[u] \arrow[r] & V \arrow[u] & S^b/\varpi^b \arrow[u] \arrow[l] \arrow[ll, bend left, "\sim"] & S^b \arrow[u] \arrow[l]
\end{tikzcd}$$

Case 2. $\varpi \neq 0$ in V $\Rightarrow$ V ϖ-torsion free, ϖ-complete w/ alg. clos. frac.

$\Rightarrow$ V perfectoid valuation ring

$$\begin{tikzcd}
S' \arrow[r] & W^\# & \exists W \arrow[l, squiggly, "\text{untilt}"'] \arrow[r, squiggly, "\text{tilt}"] & S'^b \\
S \arrow[u] \arrow[r] & V \arrow[u] & V^b \arrow[u] & S^b \arrow[l] \arrow[u]
\end{tikzcd}$$

§3.4 Coh. descent of perfectoids in arc topology

Th. (Bhatt–Scholze 2019)

Let $S \to S'$ be a ϖ-complete arc covering of perfectoid rings. Then

$$0 \to S \to S' \to S' \hat{\otimes}_S S' \to \cdots \quad \text{exact.}$$

Proof. $\Rightarrow$ $S^b \to S'^b$ ϖ^b-complete arc cover

$\Rightarrow$ $S^b \to S'^b \times S^b[\frac{1}{\varpi^b}]$ arc cover

$$0 \to S^b \to S'^b \times S^b[\tfrac{1}{\varpi^b}] \to \left(S'^b \times S^b[\tfrac{1}{\varpi^b}]\right) \otimes_{S^b} \left(S'^b \times S^b[\tfrac{1}{\varpi^b}]\right) \to \cdots \quad \text{exact}$$

$\Big\downarrow$ ϖ^b-completion

$$0 \to S^b \to S'^b \to S'^b \hat{\otimes}_{S^b} S'^b \to \cdots \quad \text{exact}$$

$$\Big\downarrow W(-)$$

$$0 \to W(S^\flat) \to W(S'^\flat) \to W(S'^\flat \hat{\otimes}_{S^\flat} S'^\flat) \to \cdots \quad \text{exact}$$

$$\xi\Big\downarrow$$

$$0 \to S \to S' \to S' \hat{\otimes}_S S' \to \cdots \quad \text{exact.}$$

<u>Def'n</u> $\forall$ ring S,

$$S_{\text{perfd}} := R\Gamma_{\substack{p\text{-complete}\\ \text{arc}}}(\operatorname{Spec} S, \mathcal{O}) \in D(S)$$

(over $\mathbb{F}_p$, $S_{\text{perfd}} = R\Gamma_{\text{arc}}(\operatorname{Spec} S, \mathcal{O})$)

<u>Rmk</u> - In general $H^q(S_{\text{perfd}}) \neq 0$ for $q > 0$.

- Th. $\Rightarrow$ if S perfd, $S = S_{\text{perfd}}$.

<u>Lecture 4</u>. <u>§4. Almost purity theorem</u>

Tate:

$$\begin{array}{ccc} L & \text{———} & \mathcal{O}_L \\ \text{finite}\,| & & | \\ \mathbb{Q}_p(\zeta_{p^\infty}) & \text{—} & \mathbb{Z}_p[\zeta_{p^\infty}] \\ \vdots & & \vdots \\ \mathbb{Q}_p(\zeta_p) & \text{—} & \mathbb{Z}_p[\zeta_p] \\ | & & | \\ \mathbb{Q}_p & \text{—} & \mathbb{Z}_p \end{array}$$

almost unramified

Faltings:

$$\begin{array}{ccc} R[\frac{1}{p}] & \text{—} & R \\ \text{finite étale}\,| & & | \text{ almost finite étale} \\ \bar{\mathbb{Q}}_p[T_1^{\pm\frac{1}{p^\infty}}, \dots, T_d^{\pm\frac{1}{p^\infty}}] & \text{—} & \bar{\mathbb{Z}}_p[T_1^{\pm\frac{1}{p^\infty}}, \dots, T_d^{\pm\frac{1}{p^\infty}}] \\ \vdots & & \vdots \\ \bar{\mathbb{Q}}_p[T_1^{\pm\frac{1}{p}}, \dots, T_d^{\pm\frac{1}{p}}] & \text{—} & \bar{\mathbb{Z}}_p[T_1^{\pm\frac{1}{p}}, \dots, T_d^{\pm\frac{1}{p}}] \\ | & & | \\ \bar{\mathbb{Q}}_p[T_1^{\pm 1}, \dots, T_d^{\pm 1}] & \text{—} & \bar{\mathbb{Z}}_p[T_1^{\pm 1}, \dots, T_d^{\pm 1}] \end{array}$$

pre-perfd.

Idea. – trivial case

$$\begin{tikzcd} \prod_{i=1}^{n} A[\frac{1}{p}] & \prod_{i=1}^{n} A \arrow[l] \\ A[\frac{1}{p}] \arrow[u, "\text{finite étale}"] & A \arrow[l] \arrow[u, "\text{finite étale}"'] \end{tikzcd}$$

integral closure

– in general, not every finite Galois $A[\frac{1}{p}]$-alg splits.

But if $A \longrightarrow B$ s.t. B is absolutely integrally closed
faithfully flat cover
of perfectoids

(i.e. any monic polynomial splits)

($\forall$ finite étale $B[\frac{1}{p}]$-alg locally splits)

we may use f-flat descent to reduce to the trivial case.

André's flatness theorem

Let A be a perfectoid ring, then $\exists$ a perfectoid A-alg B,

① B absolutely integrally closed

② $A/p^n \to B/p^n$ faithfully flat, $\forall n$

($A \to B$ p-completely faithfully flat)

§ 4.1 construct perfectoid cover by adding roots

$P \in A[T]$ monic

$A \longrightarrow A[T]/(P) \longrightarrow$ perfectoid ?

add root of P

$A \longrightarrow A[T]/(P) \longrightarrow \dfrac{A\langle T^{1/p^\infty}\rangle}{(P)}$ not perfd.

<u>Example</u>. If R perfectoid, strict pseudo-uniformizer ϖ,

$$I \subset R \quad \varpi\text{-divisible} \quad (\text{i.e. } I = \varpi I)$$

$$\widehat{R/I} = \hat{R} = R \quad \text{is perfectoid.}$$

$$A \to A[T]/(p) \to \frac{A\langle T^{1/p^\infty}\rangle[p, \frac{p}{\varpi}, \frac{p}{\varpi}, \frac{p}{\varpi^2} \cdots]}{(p, \frac{p}{\varpi}, \frac{p}{\varpi^2}, \cdots)} \subset A\langle T^{1/p^\infty}\rangle[\tfrac{1}{p}]$$

$$\begin{tikzcd}
\dfrac{A\langle T^{1/p^\infty}\rangle[\frac{p}{\varpi^{p^\infty}}]}{\varpi} \arrow[d, circlearrowleft] & \twoheadleftarrow & A\langle T^{1/p^\infty}\rangle/\varpi \arrow[d, circlearrowleft] \\
\text{Frob surj.} & \Leftarrow & \text{Frob. surj.}
\end{tikzcd}$$

<u>Lemma</u> Let A be a p-torsion free ring. $\exists\, \varpi \in pA^\times$ admits compatible p-power roots $(\varpi^{1/p^n})_n$. A^+ the p-integral closure of A in $A[\frac{1}{p}]$. Assume that Frob: $A/pA \to A/pA$ surj. then $\widehat{A^+}$ is perfectoid.

<u>Proof</u>. To eliminate $\ker\left(A/\varpi^{1/p}A \xrightarrow{\text{Frob}} A/\varpi A\right)$

$$a \longmapsto a^p \in \varpi A$$

$$a_0 = a,\ a_0^p \in \varpi A \Rightarrow \frac{a_0}{\varpi^{1/p}} \in A^+$$

$$\exists\, a_1^p \equiv a_0 \mod \varpi A \Rightarrow \left(\frac{a_1}{\varpi^{1/p^2}}\right)^p \equiv \frac{a_0}{\varpi^{1/p}} \mod \varpi A^+ \Rightarrow \frac{a_1}{\varpi^{1/p^2}} \in A^+$$

$$\cdots\cdots \quad \frac{a_2}{\varpi^{1/p^3}} \in A^+$$

We put $A_1 = A\left[\frac{a_n}{\varpi^{1/p^{n+1}}} : a \in A \text{ with } a^p \in \varpi A,\ n \geq 0\right] \subset A^+$

$\Rightarrow a \in \varpi^{1/p} A_1$ & Frob $\curvearrowright A_1/\varpi A_1$ surj.

$\Rightarrow \quad A \subset A_1 \subset A_2 \subset \cdots \quad \subset A_\infty \subset A^+$

$\Rightarrow \quad A_\infty / \varpi^{1/p} A_\infty \xrightarrow[\sim]{\text{Frob}} A_\infty / \varpi A_\infty$

$\Rightarrow$ A_∞ p-integrally closed $\Rightarrow A^+ = A_\infty$.

$\Rightarrow \hat{A}^+ = \hat{A}_\infty$ is perfectoid.

$$A \longrightarrow A[T]/(p) \longrightarrow \underbrace{\frac{\overbrace{A\langle T^{1/p^\infty}\rangle\left[\frac{p}{\varpi^\infty}\right]^+}^{\Sigma}}{\left(\frac{p}{\varpi^\infty}\right)}}_{\text{its } p\text{-completion} = \left(A\langle T^{1/p^\infty}\rangle\left[\frac{p}{\varpi^\infty}\right]^+\right)^\wedge \text{ is perfectoid.}}$$

§4.2 Structure of $A\langle T^{1/p^\infty}\rangle\left[\frac{p}{\varpi^\infty}\right]^+$

<u>Lem.</u> Let A be a p-torsion free perfectoid ring, ϖ.

$P \in A[T]$ monic poly., $m \in \mathbb{N}$. then

$$A\langle T^{1/p^\infty}\rangle\left[\frac{P}{\varpi^m}\right] \simeq A\langle T^{1/p^\infty}\rangle[X]/(\varpi^m X - P) \qquad [\text{OBIQ}]$$

$$\subset A\langle T^{1/p^\infty}\rangle\left[\tfrac{1}{p}\right]$$

is p-completely faithfully flat $/A$.

<u>Proof</u>. – Note ϖ^m, P form a regular sequence of $A\langle T^{1/p^\infty}\rangle$,

i.e. ϖ^m is nonzero divisor in $A\langle T^{1/p^\infty}\rangle$, P is nonzero divisor in $A\langle T^{1/p^\infty}\rangle/\varpi^m$.

(as P monic)

– Note $\varpi^m X - P$ is A-regular sequence

of $A\langle T^{1/p^\infty}\rangle[X]$.

i.e. $\forall$ residue field k of A, $\varpi^m X - P$ is nonzero divisor in $k[T^{1/p^\infty}][X]$.

Lemma (Special case)

If P admits compatible p^∞-roots in $A\langle T^{1/p^\infty}\rangle$, then

$$0 \to A\langle T^{1/p^\infty}\rangle\left[\frac{P}{\varpi^m}\right]^+ = A\langle T^{1/p^\infty}\rangle\left[\frac{P^{1/p^\infty}}{\varpi^{m/p^\infty}}\right].$$

$(\varpi^{m/p^\infty} x^{1/p} - P^{1/p^\infty})/\varpi^{1/p}$

Proof.

$$\begin{tikzcd}
 & \frac{A\langle T^{1/p^\infty}\rangle[x^{1/p^\infty}]}{\varpi^{1/p}} \arrow[r, two heads] \arrow[d, two heads, "\wr"] & A\langle T^{1/p^\infty}\rangle\left[\frac{P^{1/p^\infty}}{\varpi^{m/p^\infty}}\right]/\varpi^{1/p} \arrow[r] \arrow[d, two heads, "\wr"] & 0 \\
0 \arrow[r] & (\cdots)/\varpi \arrow[r] & \frac{A\langle T^{1/p^\infty}\rangle[x^{1/p^\infty}]}{\varpi} \arrow[r, two heads] & A\langle T^{1/p^\infty}\rangle\left[\frac{P^{1/p^\infty}}{\varpi^{m/p^\infty}}\right]/\varpi \arrow[r] & 0
\end{tikzcd}$$

$\wr$ Frob $\qquad$ $\wr$ Frob $\qquad$ 5. Lemma

To reduce to the special case $P^{1/p^\infty} \in A\langle T^{1/p^\infty}\rangle$, using that A is perfectoid.

(Cesnavicius–Scholze, purity for flat coh. §2)

Approximation Lemma Let A be a p-torsion free perfectoid ring, $\varpi = \xi_0^\#$

For $a \in A$ and $m \in \mathbb{N}$, there exists $b \in A^\flat$ with $b^\# = \theta([b]) \in A$ (admits p^∞-roots)

$\forall$ p-complete valuation ring $A \to V$

$$|a - b^\#|_V \le |p|_V \cdot \max\left(|b^\#|_V, |p^m|_V\right)$$

Cor. $A\left[\frac{a}{\varpi^m}\right]^+ = A\left[\frac{b^\#}{\varpi^m}\right]^+$.

Proof of Cor. $\left|\frac{a}{p^m} - \frac{b^\#}{p^m}\right|_V < \max\left(\left|\frac{b^\#}{p^m}\right|_V, 1\right)$

$$\Rightarrow \left|\frac{a}{p^m}\right|_V \le 1 \iff \left|\frac{b^\#}{p^m}\right|_V \le 1.$$

$$\Rightarrow \left|\frac{a}{p^m} - \frac{b^\#}{p^m}\right|_V < 1, \quad \forall\, A\left[\frac{a}{p^m}\right] \to V \quad \left(\iff A\left[\frac{b^\#}{p^m}\right] \to V\right)$$

$$\Rightarrow \left(\frac{a}{p^m} - \frac{b^\#}{p^m}\right)^{p^r} \in pA\left[\frac{a}{p^m}\right] \cap pA\left[\frac{b^\#}{p^m}\right] \quad \text{(exercise)}$$

$\Rightarrow \frac{a}{p^m} \in A[\frac{b^N}{p^m}]^+$, $\frac{b^N}{p^m} \in A[\frac{a}{p^m}]^+$.

<u>Proof of Lemma</u>.

$$\begin{array}{ccccc} x_0 \in & W(A^b) & \longrightarrow & A^b \ni & \overline{x_0} \\ \downarrow & \theta \downarrow & & & \\ a \in & A & & & \overline{x_0}^{\#} \end{array}$$

$$\xi = [\bar{\xi}] + p\xi' \in \ker(\theta), \quad \xi' \in W(A^b)^\times$$

We define inductively another lifting

$$x_{n+1} = x_n - \xi\xi'^{-1} x_n' \qquad \left(x_n = [\bar{x}_n] + p x_n' \right)$$

$\Rightarrow$ x_m meets our requirements.

<u>Prop</u> Let A be a p-torsion free perfectoid ring, $P \in A[T]$ monic poly.

Then $A\langle T^{1/p^\infty}\rangle[\frac{P}{\varpi^m}]^+$ is p-completely f. flat $/A$.

<u>Proof</u>. By approximation Lemma, $\exists$ $Q \in A\langle T^{1/p^\infty}\rangle$ admitting p^∞-roots, s.t

$$A\langle T^{1/p^\infty}\rangle[\frac{P}{\varpi^m}]^+ = A\langle T^{1/p^\infty}\rangle[\frac{Q}{\varpi^m}]^+. \quad \checkmark \text{ by special case.}$$

§4.3 <u>André's flatness theorem</u>

Let A be a perfectoid ring, then $\exists$ perfd A-alg B

① B absolutely integrally closed

② $A \to B$ p-completely f. flat.

<u>Proof</u>. <u>Step 1</u>. May assume A p-torsion free ($\exists$ $A' \twoheadrightarrow A$)

<u>Step 2</u>. $\forall$ $P \in A[T]$ monic,

$$A \to \left(\frac{A\langle T^{1/p^\infty}\rangle [\frac{p}{\varpi^\infty}]^+}{(\frac{p}{\varpi^\infty})} \right)^\wedge = \left(A\langle T^{1/p^\infty}\rangle [\frac{p}{\varpi^\infty}]^+ \right)^\wedge$$

<u>Step 3.</u> $A_1 = \widehat{\bigotimes_{A}}_{P \in A[T] \text{ monic}} A\langle T^{1/p^\infty}\rangle [\frac{p}{\varpi^\infty}]^+$

<u>Step 4.</u> Transfinite recursion

$$A \to A_1 \to A_2 \to \cdots \to A_{\aleph_1} = B$$

§4. <u>Almost purity theorem</u>

<u>Thm</u> (almost purity, Scholze 2013)

Let R be a perfectoid ring, S an integral R-alg s.t.

$$\begin{tikzcd} S[\frac{1}{p}] & S \arrow[l] \arrow[d, "\text{Sperfd}"'] \\ R[\frac{1}{p}] \arrow[u, "\text{f. ét}"] & R \arrow[l] \arrow[u] \end{tikzcd}$$

① $S[\frac{1}{p}]$ is finite étale $/ R[\frac{1}{p}]$

② S integrally closed in $S[\frac{1}{p}]$

Then S is perfectoid ring, almost finite étale $/ R$.

<u>Proof</u> (idea) We may assume that $S[\frac{1}{p}]$ Galois $/ R[\frac{1}{p}]$

$$\Rightarrow S[\tfrac{1}{p}] \otimes_{R[\frac{1}{p}]} S[\tfrac{1}{p}] \simeq \prod_G S[\tfrac{1}{p}]$$

$$(f, g) \longmapsto (f\sigma(g))_{\sigma \in G}$$

By Andre's flatness thm, $\exists\ R \to R'$ p-completely f. flat cover of perfectoids, with any monic R-polynomial splits in R'.

<u>Crucial case</u>: $S[\frac{1}{p}] \otimes_{R[\frac{1}{p}]} R'[\frac{1}{p}] \simeq \prod_G R'[\frac{1}{p}]$

$$\begin{tikzcd} & & S' = \prod_G R' \\ \prod_G R'[\frac{1}{p}] & S \otimes_R R' \arrow[l] \arrow[u] & \\ R'[\frac{1}{p}] \arrow[u] & R' \arrow[l] \arrow[u, "\text{integral}"'] & \end{tikzcd}$$

The integral closure S' of $S \otimes_R R'$ in $\prod_G R'[\frac{1}{p}]$ is $\prod_G R'$

$$\begin{tikzcd}
\vdots & \vdots \\
S_2 & S_1 \hat{\otimes}_R R' \\
S_1 \arrow[u] & S_0 \hat{\otimes}_R R' \arrow[u] \\
S_0 \arrow[u] \arrow[d, equal] & \\
S_{perfd} \arrow[r] & (S_{perfd} \overset{\mathbb{L}}{\otimes}_R R')^{\wedge} = (S \otimes_R R')_{perfd} \\
R \arrow[u] \arrow[r] & R' \arrow[u]
\end{tikzcd}$$

is in degree 0, almost isomorphic to S'.

$$\begin{tikzcd}
S' = \prod_G R' \arrow[r] & S'/p \\
S \otimes_R R' \arrow[u] \arrow[r] & S \otimes_R R'/p \arrow[u]
\end{tikzcd}$$

$$\Rightarrow \quad (S \otimes_R R')_{perfd} \to S' \oplus (S \otimes_R R'/p)_{perf} \to (S'/p)_{perfd}$$

$$S'/p \twoheadrightarrow S'/p^{1/p^\infty} = (S'/p)_{perfd}$$

① S_{perfd} is deg 0, and $S_{perfd}/p^n \otimes_{R/p^n} R'/p^n \xrightarrow[\sim]{almost} S'/p^n = \prod_G R'/p^n$

$\to$ S_{perfd}/p^n almost étale

$\xrightarrow{deformation}$ S_{perfd} almost étale

$$\begin{tikzcd}
S \arrow[r, "\sim"] \arrow[d] & S_{perfd} \arrow[d] \\
V \arrow[r, "\sim"] & V_{perfd}
\end{tikzcd}$$

ramification theory

<u>Lecture 5</u> §5. Galois cohomology over smooth varieties.

§5.1. Perfectoidization and Galois cohomology.

$R_{\text{perfd}} := R\Gamma_{p\text{-arc}}(\operatorname{Spec} R, \mathcal{O})$

- If R perfectoid, $R = R_{\text{perfd}}$.

- $H^q(R_{\text{perfd}})$ can be regarded as the "distance" from R to perfectoids.

How to compute it?

<u>Example</u>. $R = \overline{\mathbb{Z}}_p[T^{\pm 1}]$ $\qquad R_\infty = \overline{\mathbb{Z}}_p[T^{\pm 1/p^\infty}] = \bigoplus_{\frac{\ell}{p^n} \in \mathbb{Z}[\frac{1}{p}]} \overline{\mathbb{Z}}_p T^{\frac{\ell}{p^n}}$

$$\simeq \bigoplus_{0 \le \frac{\ell}{p^n} < 1} R T^{\frac{\ell}{p^n}}$$

faithfully flat $/R$ $\quad (\Rightarrow$ arc-covering$)$

$R_\infty / p^{\frac{1}{p}} R_\infty \xrightarrow[\sim]{\text{Frob}} R_\infty / p R_\infty$.

$\Rightarrow \widehat{R_\infty}$ is perfectoid.

$$\begin{tikzcd}
R_\infty[\frac{1}{p}] & R_\infty \arrow[l] \\
R[\frac{1}{p}] \arrow[u, "\text{pro.f.ét}"] & R \arrow[l] \arrow[u, "\text{int. clos}"']
\end{tikzcd}$$

Čech cover $\{R \to R_\infty\}$ in $\mathrm{Sch}_{p\text{-arc}}$

$$\Rightarrow R_{\text{perfd}} = \mathrm{Tot}\Big(\underset{\widehat{R_\infty}}{\underset{\shortparallel}{R_{\infty\,\text{perfd}}}} \rightrightarrows \underbrace{(R_\infty \otimes_R R_\infty)_{\text{perfd}}}_{?} \substack{\to\\\to\\\to} \underbrace{(R_\infty \otimes_R R_\infty \otimes_R R_\infty)_{\text{perfd}}}_{?} \substack{\to\\\to\\\to\\\to} \cdots\Big)$$

Fraction fields:

$$\begin{tikzcd}
K_\infty = \overline{\mathbb{Q}}_p(T^{1/p^\infty}) & \overline{\mathbb{Q}}_p[T^{\pm 1/p^\infty}] = A_\infty = R_\infty[\frac{1}{p}] \arrow[l] \\
K = \overline{\mathbb{Q}}_p(T) \arrow[u, "\text{Galois}"] & \overline{\mathbb{Q}}_p[T^{\pm 1}] = A = R[\frac{1}{p}] \arrow[l] \arrow[u, "\text{pro.f.ét}"']
\end{tikzcd}$$

$$\begin{tikzcd}
K_n = \overline{\mathbb{Q}}_p(T^{1/p^n}) & \overline{\mathbb{Q}}_p[T^{\pm 1/p^n}] = A_n \arrow[l] \\
K = \overline{\mathbb{Q}}_p(T) \arrow[u, "\text{finite Galois}", "\text{Gal gp } \mathbb{Z}/p^n\mathbb{Z}"'] & \overline{\mathbb{Q}}_p[T^{\pm 1}] = A \arrow[l] \arrow[u]
\end{tikzcd}$$

generator σ : $T^{1/p^n} \mapsto \zeta_{p^n} T^{1/p^n}$

ζ_{p^n} primitive p^n-th root of unity

Galois theory. $K_n \otimes_K K_n \xrightarrow{\sim} \prod_{\mathbb{Z}/p^n\mathbb{Z}} K_n$

$$f \otimes g \longmapsto (fg, f\sigma(g), \dots, f\sigma^{p^n-1}(g))$$

$$\begin{tikzcd}
\prod_{\mathbb{Z}/p^n\mathbb{Z}} K_n \arrow[d, equal] & \prod_{\mathbb{Z}/p^n\mathbb{Z}} A_n \arrow[l] \arrow[d, equal] & \prod_{\mathbb{Z}/p^n\mathbb{Z}} R_n \arrow[l] \\
K_n \otimes_K K_n & A_n \otimes_A A_n \arrow[l] & R_n \otimes_R R_n \arrow[l] \arrow[u, "\text{normalization}, \neq \text{id}"'] \\
K_n \arrow[u] & A_n \arrow[l] \arrow[u] & R_n \arrow[l] \arrow[u] \\
K \arrow[u, "\text{f. ét}"] & A \arrow[l] \arrow[u, "\text{f. ét}"] & R \arrow[l] \arrow[u, "\text{finite not étale}"'] \\
 & \overline{\mathbb{Q}}_p \arrow[u] & \overline{\mathbb{Z}}_p \arrow[l] \arrow[u]
\end{tikzcd}$$

not normal

$$\begin{tikzcd}
\operatorname{colim}_{n\to\infty} \operatorname{colim}_n \prod_{\mathbb{Z}/p^n\mathbb{Z}} K_\infty \arrow[d, equal] & \operatorname{colim}_n \prod_{\mathbb{Z}/p^n\mathbb{Z}} A_\infty \arrow[l] \arrow[d, equal] & \operatorname{colim}_n \prod_{\mathbb{Z}/p^n\mathbb{Z}} R_\infty \arrow[l] \\
K_\infty \otimes_K K_\infty & A_\infty \otimes_A A_\infty \arrow[l] & R_\infty \otimes_R R_\infty \arrow[l] \arrow[u, "\text{int. closure}"']
\end{tikzcd}$$

Similar as in almost purity, $\left(R_\infty \otimes_R R_\infty\right)_{\text{perfd}} \xrightarrow{\text{almost isom.}} \left(\operatorname{colim}_n \prod_{\mathbb{Z}/p^n} R_\infty\right)_{\text{perfd}}$

$$- \operatorname{colim}_n \prod_{\mathbb{Z}/p^n} R_\infty / p^{1/p} \xrightarrow[\sim]{\text{Frob}} \operatorname{colim}_n \prod_{\mathbb{Z}/p^n} R_\infty$$

$$\Rightarrow \left(\operatorname{colim}_n \prod_{\mathbb{Z}/p^n} R_\infty\right)_{\text{perfd}} = \left(\operatorname{colim}_n \prod_{\mathbb{Z}/p^n} R_\infty\right)^\wedge$$

$$= \varprojlim_r \operatorname{colim}_n \prod_{\mathbb{Z}/p^n} R_\infty / p^r R_\infty$$

$$= \varprojlim_r \operatorname{colim}_n \operatorname{Map}(\mathbb{Z}/p^n, R_\infty / p^r R_\infty)$$

$$= \varprojlim_r \operatorname{Cont}(\underbrace{\mathbb{Z}_p}_{\substack{p\text{-adic}\\ \text{top}}}, \underbrace{R_\infty / p^r R_\infty}_{\text{discrete top.}}) = \operatorname{Cont}(\mathbb{Z}_p, \hat{R}_\infty)$$

$$R_{\text{perfd}} \overset{\text{almost}}{\approx} \operatorname{Tot}\left(\hat{R}_\infty \rightrightarrows \operatorname{Cont}(\mathbb{Z}_p, \hat{R}_\infty) \substack{\rightarrow\\\rightarrow\\\rightarrow} \operatorname{Cont}(\mathbb{Z}_p \times \mathbb{Z}_p, \hat{R}_\infty) \substack{\rightarrow\\\rightarrow\\\rightarrow\\\rightarrow} \cdots\right)$$

$$= \hat{R}_\infty \longrightarrow \operatorname{Cont}(\mathbb{Z}_p, \hat{R}_\infty) \longrightarrow \operatorname{Cont}(\mathbb{Z}_p \times \mathbb{Z}_p, \hat{R}_\infty) \longrightarrow \cdots$$

$$x \longmapsto dx : \sigma \mapsto \sigma x - x$$

$$f \longmapsto df : (\sigma, \tau) \mapsto \sigma f(\tau) - f(\sigma\tau) + f(\sigma)$$

$= C^\bullet_{\text{cont}}(\mathbb{Z}_p, \hat{R}_\infty)$ the complex for group cohomology.

$= R\Gamma(\mathbb{Z}_p, \hat{R}_\infty)$

<u>Prop</u> (Faltings 1988) For

$$\begin{tikzcd}
\bar{\mathbb{Z}}_p[T_1^{\pm 1/p^\infty}, \dots, T_d^{\pm 1/p^\infty}] \arrow[r] & R_\infty \circlearrowleft \mathbb{Z}_p^d \\
\bar{\mathbb{Z}}_p[T_1^{\pm 1}, \dots, T_d^{\pm 1}] \arrow[u] \arrow[r, "\text{étale}"] \arrow[ur, phantom, "\urcorner"] & R \arrow[u]
\end{tikzcd}$$

$R_{\text{perfd}} \simeq R\Gamma(\mathbb{Z}_p^d, \hat{R}_\infty)$ in $D^{\text{al}}(R)$.

§5.2. Galois cohomology computation

Prop. For any p-complete module M endowed w/ a cts action of $\mathbb{Z}_p$, we have

$$R\Gamma(\mathbb{Z}_p, M) \simeq \left(M \xrightarrow{\sigma-1} M\right), \quad \sigma \text{ top. generator of } \mathbb{Z}_p$$

$$R = \bar{\mathbb{Z}}_p[T^{\pm 1}]$$

$\sigma-1$ acts as $\zeta_{p^n}^{l} - 1 \mid \zeta_p - 1$

$$R_\infty = \bar{\mathbb{Z}}_p[T^{\pm 1/p^\infty}] = \bigoplus_{0 \le \frac{l}{p^n} < 1} R \cdot T^{\frac{l}{p^n}} = R \oplus \underbrace{\bigoplus_{0 < \frac{l}{p^n} < 1} R \cdot T^{\frac{l}{p^n}}}_{\overset{!!}{D}}$$

$$R\Gamma(\mathbb{Z}_p, \hat{R}_\infty) = \begin{pmatrix} \hat{R} & \xrightarrow{0} & \hat{R} \\ \oplus & & \oplus \\ \hat{D} & \xrightarrow[\sigma-1]{} & \hat{D} \end{pmatrix}$$

$\sigma - 1 : D \longrightarrow D$ injective

w/ cokernel killed by $\zeta_p - 1$

$\in p^{\frac{1}{p-1}} \bar{\mathbb{Z}}_p^\times$.

$$\Rightarrow H^0(\mathbb{Z}_p, \hat{R}_\infty) \simeq \hat{R}$$

killed by $p^{\frac{1}{p-1}}$

$$H^1(\mathbb{Z}_p, \hat{R}_\infty) \simeq \hat{R} \oplus \hat{D}/(\sigma-1)\hat{D}$$

$$\Rightarrow H^0(\mathbb{Z}_p, \hat{R}_\infty[\tfrac{1}{p}]) \simeq \hat{R}[\tfrac{1}{p}]$$

$$H^1(\mathbb{Z}_p, \hat{R}_\infty[\tfrac{1}{p}]) \simeq \hat{R}[\tfrac{1}{p}]$$

Prop. For $\bar{\mathbb{Z}}_p[T_1^{\pm 1}, \dots, T_d^{\pm 1}] \xrightarrow{\text{étale}} R$,

$$H^q(\mathbb{Z}_p^d, \hat{R}_\infty[\tfrac{1}{p}]) \simeq \Lambda^q\left(\hat{R}[\tfrac{1}{p}]^{\oplus d}\right)$$

$$\wr\wr \qquad\qquad \wr\wr$$

$$H^q(G_R, \hat{R}[\tfrac{1}{p}]) \overset{\text{canonical}}{\simeq} \hat{\Omega}^q_{R[\frac{1}{p}]/\mathbb{Z}_p}$$

$$\begin{tikzcd}
\bar{R}[\frac{1}{p}] & \bar{R} \arrow[l] \\
R[\frac{1}{p}] \arrow[u, "\text{univ. cover}"] \arrow[d, bend right] & R_\infty[\frac{1}{p}] \arrow[l] \arrow[u] \\
\pi_1(R[\frac{1}{p}]) =: G_R & R \arrow[l] \arrow[u] \arrow[ur, "\text{int. clos.}"] \arrow[ur, bend right, "R_\infty"']
\end{tikzcd}$$

§5.3 Faltings extension

Th. (Faltings) There is a canonical exact sequence (independent of the chart)

$$0 \to \hat{\bar{R}}[\tfrac{1}{p}] \to F_R \to \hat{\bar{R}}[\tfrac{1}{p}] \otimes_R \Omega^1_{R/\bar{\mathbb{Z}}_p}(-1) \to 0$$

s.t. $G_R = \pi_1(R[\tfrac{1}{p}])$-inv.

$$0 \to \hat{\bar{R}}[\tfrac{1}{p}] \xrightarrow{\sim} F_R^{G_R} \xrightarrow{0} \overbrace{\hat{\bar{R}}[\tfrac{1}{p}] \otimes_R \Omega^1_{R/\bar{\mathbb{Z}}_p}(-1)}^{\hat{\Omega}^1_{R[\frac{1}{p}]/\bar{\mathbb{Z}}_p}(-1)} \xrightarrow[\sim]{\delta} H^1(G_R, \hat{\bar{R}}[\tfrac{1}{p}])$$

Construction. (simplification

$$\begin{array}{ccc} R_{\bar{\mathbb{Z}}_p} & - & R \\ | & & | \text{ smooth} \\ \bar{\mathbb{Z}}_p & - & \mathbb{Z}_p \end{array}$$

$$\begin{array}{ccccc} \bar{R}_{\bar{\mathbb{Q}}_p} & - & \bar{R} & & \\ G_R \,| & & | & & \\ R_{\bar{\mathbb{Q}}_p} & - & R_{\bar{\mathbb{Z}}_p} & - & R \\ | & & | & & | \text{ smooth} \\ \bar{\mathbb{Q}}_p & - & \bar{\mathbb{Z}}_p & - & \mathbb{Z}_p \end{array}$$

$$\Omega^1_{R/\mathbb{Z}_p} \otimes_R \bar{R} \to \Omega^1_{\bar{R}/\mathbb{Z}_p} \to \Omega^1_{\bar{R}/R} \to 0$$

$$\wr\wr \quad \bar{R}^{\oplus d}$$

$$\underbrace{\Omega^1_{\bar{\mathbb{Z}}_p/\mathbb{Z}_p} \otimes_{\bar{\mathbb{Z}}_p} \bar{R}}_{\wr\wr \text{ almost}} \rightleftarrows \Omega^1_{\bar{R}/\mathbb{Z}_p} \to \Omega^1_{\bar{R}/\bar{\mathbb{Z}}_p} \to 0$$

$$\wr\wr \quad \bar{R}[\tfrac{1}{p}]/p^{-\frac{1}{p-1}}\bar{R} \oplus \bar{R}[\tfrac{1}{p}]^{\oplus d}$$

almost unram.

$$\begin{array}{c} \bar{\mathbb{Z}}_p \\ | \\ \mathbb{Z}_p[\zeta_{p^\infty}] \\ | \\ \mathbb{Z}_p \end{array}$$

$$\Omega^1_{\mathbb{Z}_p[\zeta_{p^\infty}]/\mathbb{Z}_p} \otimes \bar{R} \qquad d\log \zeta_{p^n} = \frac{d\zeta_{p^n}}{\zeta_{p^n}}$$

$$\simeq \bar{R}[\tfrac{1}{p}]/p^{-\frac{1}{p-1}}\bar{R} \qquad p^{-n}$$

$$\begin{array}{c} \bar{R} \\ | \\ (R_{\bar{\mathbb{Z}}_p})_\infty \\ | \\ \bar{\mathbb{Z}}_p \end{array} \overset{\text{almost unram.}}{\sim} \bar{\mathbb{Z}}_p[T_1^{\pm 1/p^\infty}, \dots, T_d^{\pm 1/p^\infty}]$$

$$\Omega^1_{\bar{R}/\bar{\mathbb{Z}}_p} \overset{\text{almost}}{\simeq} \Omega^1_{(R_{\bar{\mathbb{Z}}_p})_\infty/\bar{\mathbb{Z}}_p} \otimes \bar{R} \simeq \bar{R}[\tfrac{1}{p}]^{\oplus d}$$

$$d\log T_i^{1/p^n} \leftarrow\!\shortmid p^{-n} e_i$$

almost exact sequence

$$\begin{tikzcd}
0 \arrow[r] & \Omega^1_{R/\mathbb{Z}_p} \otimes_R \bar{R} \arrow[r] & \Omega^1_{\bar{R}/\mathbb{Z}_p} \arrow[r] & \Omega^1_{\bar{R}/R} \arrow[r] & 0 \\
0 \arrow[r] & \bar{R}^{\oplus d} \arrow[r] & \bar{R}[\tfrac{1}{p}]/p^{-\frac{1}{p-1}}\bar{R} \oplus \bar{R}[\tfrac{1}{p}]^{\oplus d} \arrow[r] & \bar{R}[\tfrac{1}{p}]/p^{-\frac{1}{p-1}}\bar{R} \oplus \left(\bar{R}[\tfrac{1}{p}]/\bar{R}\right)^{\oplus d} \arrow[r] & 0
\end{tikzcd}$$

$$\begin{tikzcd}
\mathrm{d}\log T_i & \mathrm{d}\log \zeta_{p^n} & \mathrm{d}\log T_i^{1/p^n} & \mathrm{d}\log \zeta_{p^n} & \mathrm{d}\log T^{1/p^n} \\
e_i \arrow[u, mapsto] \arrow[r, phantom, "\cong"] & p^{-n}e_0 \arrow[u, mapsto] \arrow[r, phantom, "\cong"] & p^{-n}e_i \arrow[u, mapsto] \arrow[r, phantom, "\cong"] & p^{-n}e_0 \arrow[u, mapsto] & p^{-n}e_i \arrow[u, mapsto]
\end{tikzcd}$$

$$\xrightarrow{R\mathrm{Hom}(\mathbb{Z}/p^n,\,-)} \quad 0 \to \underbrace{\Omega^1_{R/\mathbb{Z}_p} \otimes_R \bar{R}[p^n]}_{\cong\, 0} \to \underbrace{\Omega^1_{\bar{R}/\mathbb{Z}_p}[p^n]}_{\cong\, p^{n-\frac{1}{p-1}}\bar{R}/p^{-\frac{1}{p-1}}\bar{R}} \to \underbrace{\Omega^1_{\bar{R}/R}[p^n]}_{\cong\, 0} \to$$

$$\Omega^1_{R/\mathbb{Z}_p} \otimes_R \bar{R}/p^n \to \Omega^1_{\bar{R}/\mathbb{Z}_p}/p^n \to \Omega^1_{\bar{R}/R}/p^n \to 0$$

$$\Omega^1_{R/\mathbb{Z}_p} \otimes_R \bar{R}/p^n \cong (\bar{R}/p^n)^{\oplus d}, \qquad \Omega^1_{\bar{R}/R}/p^n \cong 0$$

$$\xrightarrow{R\lim} \quad 0 \to \hat{\bar{R}}_{[\frac{1}{p}]}(\mathrm{d}\log \zeta_{p^n})_{n\in\mathbb{N}} \to \left(\varprojlim_n \Omega^1_{\bar{R}/R}[p^n]\right)^{[\frac{1}{p}]} \to \hat{\bar{R}}_{[\frac{1}{p}]} \otimes_R \Omega^1_{R/\mathbb{Z}_p} \to 0$$

$$\hat{\bar{R}}_{[\frac{1}{p}]}(\mathrm{d}\log \zeta_{p^n})_{n\in\mathbb{N}} \cong \hat{\bar{R}}[\tfrac{1}{p}](1)$$

$$(-1)\Big\downarrow$$

$$0 \to \hat{\bar{R}}[\tfrac{1}{p}] \to F_R \to \hat{\bar{R}}[\tfrac{1}{p}] \otimes_R \Omega^1_{R/\mathbb{Z}_p}(-1) \to 0.$$

§5.4 Hodge-Tate decomposition

<u>Faltings' Galois cohomology computation</u>: For $R/\bar{\mathbb{Z}}_p$ smooth, s.t. admit a chart $\bar{\mathbb{Z}}_p[T_1^{\pm 1}, \dots, T_d^{\pm 1}] \xrightarrow{\text{étale}} R$, there is a canonical isom.

$$\hat{\Omega}^q_{R[\frac{1}{p}]/\bar{\mathbb{Q}}_p}(-q) \xrightarrow{\sim} H^q(R_{\mathrm{perfd}})[\tfrac{1}{p}] = H^q_{p\text{-arc}}(\mathrm{Spec}\, R, \mathcal{O})[\tfrac{1}{p}]$$

<u>Proof</u>. §5.2, 5.3.

<u>Faltings' main comparison theorem</u>. For $X/\overline{\mathbb{Z}}_p$ proper smooth,

$$H^q_{\acute{e}t}(X_{\overline{\mathbb{Q}}_p}, \mathbb{C}_p) \xrightarrow{\sim} H^q_{p\text{-arc}}(X, \mathcal{O})[\tfrac{1}{p}].$$

<u>Idea</u>. <u>Step 1</u>. $H^q_{\acute{e}t}(X_{\overline{\mathbb{Q}}_p}, \mathbb{Z}/p^n) \simeq H^q_{p\text{-arc}}(X, \mathbb{Z}/p^n)$, $\forall X$

by reducing to val. ring.

<u>Step 2</u> Almost finiteness (X proper)

$H^q_{p\text{-arc}}(X, \mathcal{O}/p) \overset{a}{\simeq} (\mathcal{O}_{\mathbb{C}_p}/p)^r$, using local Galois coh.

$\Rightarrow H^q_{p\text{-arc}}(X, \mathcal{O}^\flat/p^\flat) \overset{a}{\simeq} (\mathcal{O}_{\mathbb{C}_p^\flat}/p^\flat)^r$

$\overset{\text{Lem}}{\Longrightarrow} H^q_{p\text{-arc}}(X, \mathcal{O}^\flat) \overset{a}{\simeq} (\mathcal{O}_{\mathbb{C}_p^\flat})^r$

$\Rightarrow H^q_{p\text{-arc}}(X, \mathcal{O}^\flat[\tfrac{1}{p^\flat}]) \simeq (\mathbb{C}_p^\flat)^r$.

<u>Step 3</u>. Artin-Schreier $0 \to \mathbb{F}_p \to \mathcal{O}^\flat[\tfrac{1}{p^\flat}] \xrightarrow{x^p - x} \mathcal{O}^\flat[\tfrac{1}{p^\flat}] \to 0$

$\Rightarrow H^q_{p\text{-arc}}(X, \mathbb{F}_p) = H^q_{p\text{-arc}}(X, \mathcal{O}^\flat[\tfrac{1}{p^\flat}])^{\mathrm{Frob}=1} \simeq \mathbb{F}_p^r$

<u>Hodge-Tate decomposition</u> For $X/\overline{\mathbb{Z}}_p$ proper smooth,

$$H^n_{\acute{e}t}(X_{\overline{\mathbb{Q}}_p}, \mathbb{C}_p) \simeq \bigoplus_{i+j=n} H^i(X, \Omega^j_{X/\overline{\mathbb{Z}}_p}(-j)) \otimes_{\overline{\mathbb{Z}}_p} \mathbb{C}_p$$

<u>Idea</u>.

$$\begin{array}{c} (\mathrm{Sch}/X)_{p\text{-arc}} \\ \downarrow \nu \\ (\mathrm{Sch}/X)_{\mathrm{Zar}} \end{array}$$

$R^q\nu_* \mathcal{O}[\tfrac{1}{p}] \simeq \hat{\Omega}^q(-q)$

$H^i_{\mathrm{Zar}}(X, \hat{\Omega}^j(-j)) \Rightarrow H^{i+j}_{p\text{-arc}}(X, \mathcal{O}[\tfrac{1}{p}]) \simeq H^{i+j}_{\acute{e}t}(X, \mathbb{C}_p)$

$$\xrightarrow[\text{Tate}]{\mathrm{Gal}(\overline{\mathbb{Q}}_p|\mathbb{Q}_p)} \bigoplus_{i+j=n} H^i(X, \hat{\Omega}^j(-j)) \simeq H^{i+j}_{\acute{e}t}(X, \mathbb{C}_p)$$

Lecture 6 §6. Galois cohomology over valuation rings

Last time.

$$\begin{array}{c} R \\ \uparrow \text{ étale} \\ \bar{\mathbb{Z}}_p[T_1^{\pm 1}, \dots, T_d^{\pm 1}] \end{array}$$

$$H^q(R_{\text{perfd}})[\tfrac{1}{p}] = H^q(G_R, \hat{\bar{R}}[\tfrac{1}{p}])$$

geom fundamental group of $R[\frac{1}{p}]$

arc-descent of perfd.

$$\overset{\text{Faltings}}{\underset{\text{extension}}{\simeq}} \hat{\Omega}^q_{R[\frac{1}{p}]/\bar{\mathbb{Q}}_p}(-q)$$

$$\Rightarrow \quad \begin{array}{c} X \\ | \\ \bar{\mathbb{Z}}_p \end{array} \text{ proper smooth} \qquad H^n(X_{\text{perfd}})[\tfrac{1}{p}] \simeq \bigoplus_{i+j=n} H^i(X, \Omega^j_{X/\bar{\mathbb{Q}}_p}(-j))$$

$$\| $$

$$H^n_{\text{ét}}(X_{\bar{\mathbb{Q}}_p}, \mathbb{C}_p)$$

But sometimes, we may encounter schemes of infinite type, but not pro-smooth.

Today: understand ramification over general (non-discrete) valuation rings.

Difficulty: $R / \bar{\mathbb{Z}}_p$ smooth $\qquad \mathcal{O}_K / \bar{\mathbb{Z}}_p$ val. ring

$R_\infty = R[T_1^{\pm 1/p^\infty}, \dots, T_d^{\pm 1/p^\infty}]$ $\qquad K_\infty = K(t_1^{1/p^\infty}, \dots, t_d^{1/p^\infty})$. $t_1, \dots, t_d$?

$\mathcal{O}_{K_\infty} = ?$

§6.1. Different ideals and differentials

$$\begin{array}{c} L \\ | \\ K \end{array}$$ finite separable ext'n of non-discrete henselian valuation fields of rank 1 (i.e. $\dim \mathcal{O}_K = \dim \mathcal{O}_L = 1$)

$\Downarrow$

$v_K : K^\times \to \mathbb{R}$ with dense image

$m_K^2 = m_K$, $m_L = m_K \mathcal{O}_L$.

$\mathrm{Tr}_{L|K} : L \longrightarrow K$

$x \longmapsto$ trace of K-linear map $L \xrightarrow{\times x} L$

$$\left(= \sum_{\sigma : L \hookrightarrow \bar{K}} \sigma(x) \right)$$

induces perfect pairing $L \times L \longrightarrow K$, i.e. $L \xrightarrow{\sim} \mathrm{Hom}_K(L, K)$

$(x, y) \mapsto \mathrm{Tr}(xy)$ $\qquad x \longmapsto \left[y \mapsto \mathrm{Tr}_{L|K}(xy) \right]$

$\mathcal{O}_L^* := \{ x \in L : \mathrm{Tr}_{L|K}(xy) \in \mathcal{O}_K, \ \forall y \in \mathcal{O}_L \}$ codifferent of $L|K$

$\mathcal{D}_{L|K} := \mathcal{O}_L^{*-1} = \{ x \in \mathcal{O}_L : x\mathcal{O}_L^* \subset \mathcal{O}_L \}$ different of $L|K$

$\Rightarrow \mathfrak{m}_L \subset \mathcal{O}_L^* \cdot \mathcal{D}_{L|K} \subset \mathcal{O}_L$

<u>Relation with trace</u>. For any nonzero fractional ideal $\mathfrak{a}$ of $\mathcal{O}_K$, $\mathfrak{b}$ of $\mathcal{O}_L$

$$\mathrm{Tr}_{L|K}(\mathfrak{m}_L \mathfrak{b}) \subset \mathfrak{a} \iff \mathfrak{b} \cdot \mathcal{D}_{L|K} \subset \mathfrak{a}\mathcal{O}_L$$

<u>Cor</u>. $|\mathrm{Tr}_{L|K}(x)| \leq |\mathcal{D}_{L|K}| \cdot |x|, \quad \forall x \in L$

<u>Idea</u>. Since the valuations on L & K are non-discrete, the test ideals $\mathfrak{a}$ & $\mathfrak{b}$ are dense in $\mathbb{R}$.

<u>Cor</u>. $(x = 1)$ $\quad |[L:K]| \leq |\mathcal{D}_{L|K}| \leq 1$ $\qquad$ (not true for discrete)

<u>Relation with differentials</u>

<u>Th</u>. (Gabber - Ramero) $\Omega^1_{\mathcal{O}_L/\mathcal{O}_K}$ is uniformly almost finitely gen. $/\mathcal{O}_L$

(i.e. $\exists\, n \in \mathbb{N}$, $\forall \varepsilon \in \mathfrak{m}_L$, $\exists\, \mathcal{O}_L^{\oplus n} \xrightarrow{f_\varepsilon} \Omega^1_{\mathcal{O}_L/\mathcal{O}_K}$ with coker killed by ε)

and $\mathcal{D}_{L|K} \overset{\text{almost}}{\simeq} \prod_{q=1}^{\infty} \mathrm{Ann}_{\mathcal{O}_L}\left(\Omega^{q}_{\mathcal{O}_L/\mathcal{O}_K}\right)$

$\rightarrow m_L \mathcal{D}_{L|K} \subset \mathrm{Ann}_{\mathcal{O}_L}\left(\Omega^{1}_{\mathcal{O}_L/\mathcal{O}_K}\right)$

<u>Idea</u>. explicit understanding of structure of $\mathcal{O}_L$.

<u>Th.</u> (Gabber–Ramero) (1) $\mathbb{L}_{\mathcal{O}_L/\mathcal{O}_K} = \Omega^1_{\mathcal{O}_L/\mathcal{O}_K}$ for $L|K$ finite separable

(2) $\Omega^1_{\mathcal{O}_K|\overline{\mathbb{Z}}_p}$ torsion free.

§6-2 Structure of $\Omega^1_{\mathcal{O}_K/\overline{\mathbb{Z}}_p}$ and control of $\mathcal{D}_{K_\infty|K}$.

A priori, $\Omega^1_{\mathcal{O}_K/\overline{\mathbb{Z}}_p}$ is torsion-free $\mathcal{O}_K$-module $\subset \Omega^1_{\mathcal{O}_K/\overline{\mathbb{Z}}_p}[\frac{1}{p}] = \Omega^1_{K/\overline{\mathbb{Q}}_p}$

of dimension tr. deg $K|\overline{\mathbb{Q}}_p$ (suppose $< +\infty$).

<u>Th.</u> (H. 2025) Let M be a torsion free $\mathcal{O}_K$-module w/ $\dim_K M[\frac{1}{p}] < \infty$, then

$$0 \to \hat{K}^{\oplus r} \to M \otimes_{\mathcal{O}_K} \mathcal{O}_{\hat{K}} \to \hat{M} \to 0.$$

Moreover, $\hat{M}$ is almost finite free over $\mathcal{O}_{\hat{K}}$.

<u>Cor.</u> Suppose tr. deg $(K|\overline{\mathbb{Q}}_p) < \infty$. $\left(\begin{array}{c} M = \Omega^1_{\mathcal{O}_K/\overline{\mathbb{Z}}_p} \\ K = \hat{\overline{K}} \end{array}\right)$

(1) $\Omega^1_{\mathcal{O}_K/\overline{\mathbb{Z}}_p} \otimes_{\mathcal{O}_K} \mathcal{O}_{\hat{K}} \twoheadrightarrow \hat{\Omega}^1_{\mathcal{O}_K/\overline{\mathbb{Z}}_p}$

(2) $\exists\, d \in \mathbb{N}$ ($\leq$ tr.deg) s.t. $\forall\, \varepsilon \in \mathfrak{m}_K$, $\exists\, t_1, \dots, t_d \in \mathcal{O}_K^\times$ s.t.

$$\varepsilon \hat{\Omega}^1_{\mathcal{O}_K/\overline{\mathbb{Z}}_p} \subset \mathcal{O}_{\hat{K}}\, dt_1 \oplus \cdots \oplus \mathcal{O}_{\hat{K}}\, dt_d \subset \hat{\Omega}^1_{\mathcal{O}_K/\overline{\mathbb{Z}}_p}.$$

For simplicity, we assume that $d=1$. $\forall \varepsilon \in m_K$, $\exists\, t \in \mathcal{O}_K^\times$ s.t.
(i.e. $|p| < |\varepsilon| < |1|$)

$$\varepsilon \hat{\Omega}^1_{\mathcal{O}_K/\mathbb{Z}_p} \subset \mathcal{O}_{\hat{K}}\, dt \subset \hat{\Omega}^1_{\mathcal{O}_K/\mathbb{Z}_p}$$

Want to understand the ramification of $K_n = K(t^{1/p^n}) \,|\, K$

At least, $\mathcal{O}_K[t^{1/p^n}] \subset \mathcal{O}_{K_n}$ (expect they are close)

<u>Lemma</u> $dt^{1/p^n} \in \Omega^1_{\mathcal{O}_{K_n}/\mathcal{O}_K}$

$$\mathrm{Ann}_{\mathcal{O}_{K_n}}(dt^{1/p^n})$$

$$|p^n| \le |\mathrm{Ann}_{\mathcal{O}_{K_n}}(dt^{1/p^n})| \le |p^n/\varepsilon|$$

<u>Proof.</u>

$$H_1(\mathbb{L}_{\mathcal{O}_{K_n}/\mathcal{O}_K}) = 0 \to \mathcal{O}_{K_n} \otimes_{\mathcal{O}_K} \Omega^1_{\mathcal{O}_K/\mathbb{Z}_p} \xrightarrow{f} \Omega^1_{\mathcal{O}_{K_n}/\mathbb{Z}_p} \to \Omega^1_{\mathcal{O}_{K_n}/\mathcal{O}_K} \to 0$$

$$\exists\, \omega \qquad\qquad dt^{1/p^n}$$

Suppose $p^n/\varepsilon' \in \mathrm{Ann}_{\mathcal{O}_{K_n}}(dt^{1/p^n})$ $\to$ $p^n/\varepsilon' \cdot dt^{1/p^n} = f(\omega)$
$(|\varepsilon'| < |\varepsilon|)$

$$\Rightarrow f(dt) = p^n t^{\frac{p^n-1}{p^n}} dt^{\frac{1}{p^n}} = \varepsilon' t^{\frac{p^n-1}{p^n}} f(\omega)$$

$$\Rightarrow dt \in \varepsilon' \Omega^1_{\mathcal{O}_K/\mathbb{Z}_p} \quad \text{contradiction.}$$

<u>Prop.</u> $|p^n| \le |D_{K_n|K}| \le |p^n/\varepsilon|$.

<u>Proof.</u> $m_{K_n} D_{K_n|K} \subset \mathrm{Ann}_{\mathcal{O}_{K_n}}(\Omega^1_{\mathcal{O}_{K_n}/\mathcal{O}_K}) \subset \mathrm{Ann}_{\mathcal{O}_{K_n}}(dt^{1/p^n}) \overset{\text{Lemma}}{\subset} \frac{p^n}{\varepsilon}\mathcal{O}_{K_n}$

<u>§6.3 Structure of $\mathcal{O}_{K_\infty}$</u>

<u>Cor.</u> $|p^{-n}\, \mathrm{Tr}_{K_n|K}(x)| \le |x/\varepsilon|$, $\forall x \in K_n$

Prop. $\varepsilon \mathcal{O}_{K_n} \subset \mathcal{O}_K[t^{1/p^n}] \subset \mathcal{O}_{K_n}$.

Proof. $\forall\, x \in \varepsilon \mathcal{O}_{K_n}$, we write $x = a_0 + a_1 t^{1/p^n} + \cdots + a_{p^n-1}\, t^{\frac{p^n-1}{p^n}}$, $a_j \in K$.

Since $\mathrm{Tr}_{K_n|K}\left(a t^{\frac{j}{p^n}}\right) = \sum_{i=0}^{p^n-1} \left(\zeta_{p^n}^{i}\right)^{j} a t^{\frac{j}{p^n}} = \begin{cases} p^n a & , j=0 \\ 0 & , j \neq 0 \end{cases}$

we have $\mathrm{Tr}_{K_n|K}\left(x t^{-\frac{j}{p^n}}\right) = p^n a_j$

$$\Rightarrow |a_j| = \left| p^{-n} \mathrm{Tr}_{K_n|K}\left(x t^{-\frac{j}{p^n}}\right)\right| \leq \left| x t^{-\frac{j}{p^n}} / \varepsilon \right| = |x/\varepsilon| \leq 1.$$

Cor. $\varepsilon \mathcal{O}_{K_\infty} \subset \mathcal{O}_K[t^{1/p^\infty}] \subset \mathcal{O}_{K_\infty}$.

Th. (H. 2025) For any henselian valuation field K of rank 1 extension of $\mathbb{Q}_p$ w/ tr. deg $< \infty$. then there exists $d \leq \text{tr. deg}(K|\mathbb{Q}_p)$ s.t. $\forall\, \varepsilon \in \mathfrak{m}_K$, $\exists\, t_1, \ldots, t_d \in \mathcal{O}_K^\times$ s.t. for $K_\infty = K(t_1^{1/p^\infty}, \ldots, t_d^{1/p^\infty})$, $\varepsilon \mathcal{O}_{K_\infty} \subset \mathcal{O}_K[t^{1/p^\infty}, \ldots, t_d^{1/p^\infty}] \subset \mathcal{O}_{K_\infty}$.

Moreover, $\mathcal{O}_{K_\infty}$ is pre-perfectoid.

Cor. $H^q(\mathbb{Z}_p^d, \hat{K}_\infty) \simeq \Lambda^q(\hat{K}^{\oplus d})$ where $\mathrm{Gal}(K_\infty|K) \simeq \mathbb{Z}_p^d$.

Proof is similar to smooth case. $\square$

$$H^q(G_K, \hat{\bar{K}}) \underset{\text{canonical}}{\overset{?}{\simeq}} \hat{\Omega}^q_{\mathcal{O}_K/\mathbb{Z}_p}[\tfrac{1}{p}]$$

§6.4. Faltings extension.

Th. (H. 2024) There is a canonical exact sequence

$$0 \to \hat{\bar{K}} \to F_K \to \hat{\bar{K}} \otimes_K \Omega^1_{K|\mathbb{Q}_p}(-1) \to 0$$

s.t.

$$\begin{tikzcd}
0 \arrow[r] & \hat{\bar{K}} \arrow[r] & F_K^{G_K} \arrow[r] & \hat{\bar{K}} \otimes_K \Omega^1_{K|\mathbb{Q}_p}(-1) \arrow[r, "\delta"] \arrow[d, two heads] & H^1(G_K, \hat{\bar{K}}) \\
 & & & \hat{\Omega}^1_{K|\mathbb{Q}_p}(-1) \arrow[ur, hook] &
\end{tikzcd}$$

G_K-inv.

§6.5 Perfectoidness criterion

$$\begin{tikzcd}
 & & K & \mathcal{O}_K \arrow[l] \\
\widetilde{R}[\tfrac{1}{p}] \arrow[urr, "\text{res}"] & \widetilde{R} \arrow[l] \arrow[ur, "\text{int. clos.}"'] & & \\
R[\tfrac{1}{p}] \arrow[u, "\text{perf. ét}"] & R \arrow[l] \arrow[u] & & \\
\overline{\mathbb{Q}}_p \arrow[u] & \overline{\mathbb{Z}}_p \arrow[l] \arrow[u] & &
\end{tikzcd}$$

Wonder: is there a simple condition of $\widetilde{R}[\frac{1}{p}]$ that implies every $\mathcal{O}_K$ pre-perfectoid?

<u>Example</u> $\widetilde{R}[\frac{1}{p}]$ abs. int. closed

$\Rightarrow K = \overline{K}$

$\Rightarrow \mathcal{O}_K / p^{1/p}\mathcal{O}_K \xrightarrow[\sim]{\text{Frob}} \mathcal{O}_K / p\mathcal{O}_K$

$\Rightarrow \widehat{\mathcal{O}_K}$ perfd.

<u>Cor.</u> If $\dim_{\widehat{K}} \mathcal{F}_K^{G_K} = 1 + \operatorname{tr.deg}(K \mid \overline{\mathbb{Q}}_p)$, then $\mathcal{O}_{\widehat{\overline{K}}}$ perfectoid

<u>Proof.</u> $\Rightarrow \delta = 0 \Rightarrow \Omega^1_{\mathcal{O}_K/\overline{\mathbb{Z}}_p} = 0 \Rightarrow \Omega^1_{\mathcal{O}_K/\overline{\mathbb{Z}}_p} = p\,\Omega^1_{\mathcal{O}_K/\overline{\mathbb{Z}}_p}$

$\Omega^1_{\mathcal{O}_{\overline{K}}/\mathcal{O}_K}[p^n] = 0$

$$\begin{tikzcd}
0 \arrow[r] & \Omega^1_{\mathcal{O}_K/\overline{\mathbb{Z}}_p} \otimes_{\mathcal{O}_K} \mathcal{O}_{\overline{K}} \arrow[r] \arrow[d, "p^n"] & \Omega^1_{\mathcal{O}_{\overline{K}}/\overline{\mathbb{Z}}_p} \arrow[r] \arrow[d, "p^n"] & \Omega^1_{\mathcal{O}_{\overline{K}}/\mathcal{O}_K} \arrow[r] \arrow[d, "p^n"] & 0 \\
0 \arrow[r] & \Omega^1_{\mathcal{O}_K/\overline{\mathbb{Z}}_p} \otimes_{\mathcal{O}_K} \mathcal{O}_{\overline{K}} \arrow[r] \arrow[d] & \Omega^1_{\mathcal{O}_{\overline{K}}/\overline{\mathbb{Z}}_p} \arrow[r] \arrow[d] & \Omega^1_{\mathcal{O}_{\overline{K}}/\mathcal{O}_K} \arrow[r] & 0 \\
 & 0 & 0 & &
\end{tikzcd}$$

$\Rightarrow \Omega^1_{\mathcal{O}_{\overline{K}}/\mathcal{O}_K} = 0 \Rightarrow \mathcal{O}_{\overline{K}} \mid \mathcal{O}_K$ almost weakly étale ... $\square$

§6.6 Application to Shimura varieties

(G, X) Shimura data, $E \mid \mathbb{Q}$ reflex field

$K^p \subset G(\mathbb{A}_f^p)$ compact open.

$(\mathrm{Sh}_{K^p K_p})_{K_p \subset G(\mathbb{Q}_p)}$ inverse system of smooth E-varieties w/ finite étale transition.

$Sh_{K^p} := \varprojlim_{K_p \to 1} Sh_{K^p K_p}$ Shimura variety at infinite level K^p.

<u>Calegari-Emerton Conj.</u> $H^q_{\acute{e}t}(Sh_{K^p, \mathbb{C}}, \mathbb{Z}_p) = 0$, $\forall\ q > \dim Sh_{K^p}$.

First breakthrough: Scholze (2013) for (G, X) of Hodge type,

Sh_{K^p} perfectoid space

$$\text{étale coh. in } \mathbb{Z}_p \xleftrightarrow{\text{Faltings}} p\text{-arc coh. in } \mathcal{O}$$

$$\xleftrightarrow[\text{perfd}]{} \text{Zariski coh. in } \mathcal{O}.$$

<u>Question</u> Is Sh_{K^p} perfectoid?

<u>Th.</u> (H. 2024) In general, Sh_{K^p} is pointwise perfectoid.

$\forall$ residue field k of $Sh_{K^p, \overline{\mathbb{Q}}_p}$,
any valuation ring of k ext. of $\overline{\mathbb{Z}}_p$ is pre-perfectoid.

<u>Proof.</u> Using our perfd criterion derived from p-adic Hodge theory for valuation rings.

<u>Cor.</u> Calegari-Emerton Conj. is true.